CNENGR202 Instrumentation and Control

Part 3 L6 – PWM - I

Prof. Zhang Liwei 张立伟 教授

Email: lwzhang@bjtu.edu.cn
Working Room: SX702

School of Electronics and Information Engineering
Beijing Jiaotong University

Recap

- Interrupts are important!
- Upon interrupt, the CPU attempts to jump to a predefined program location defined for handling the particular interrupt
- Internal interrupt
- External interrupt

Set MAX

输出比较寄存器

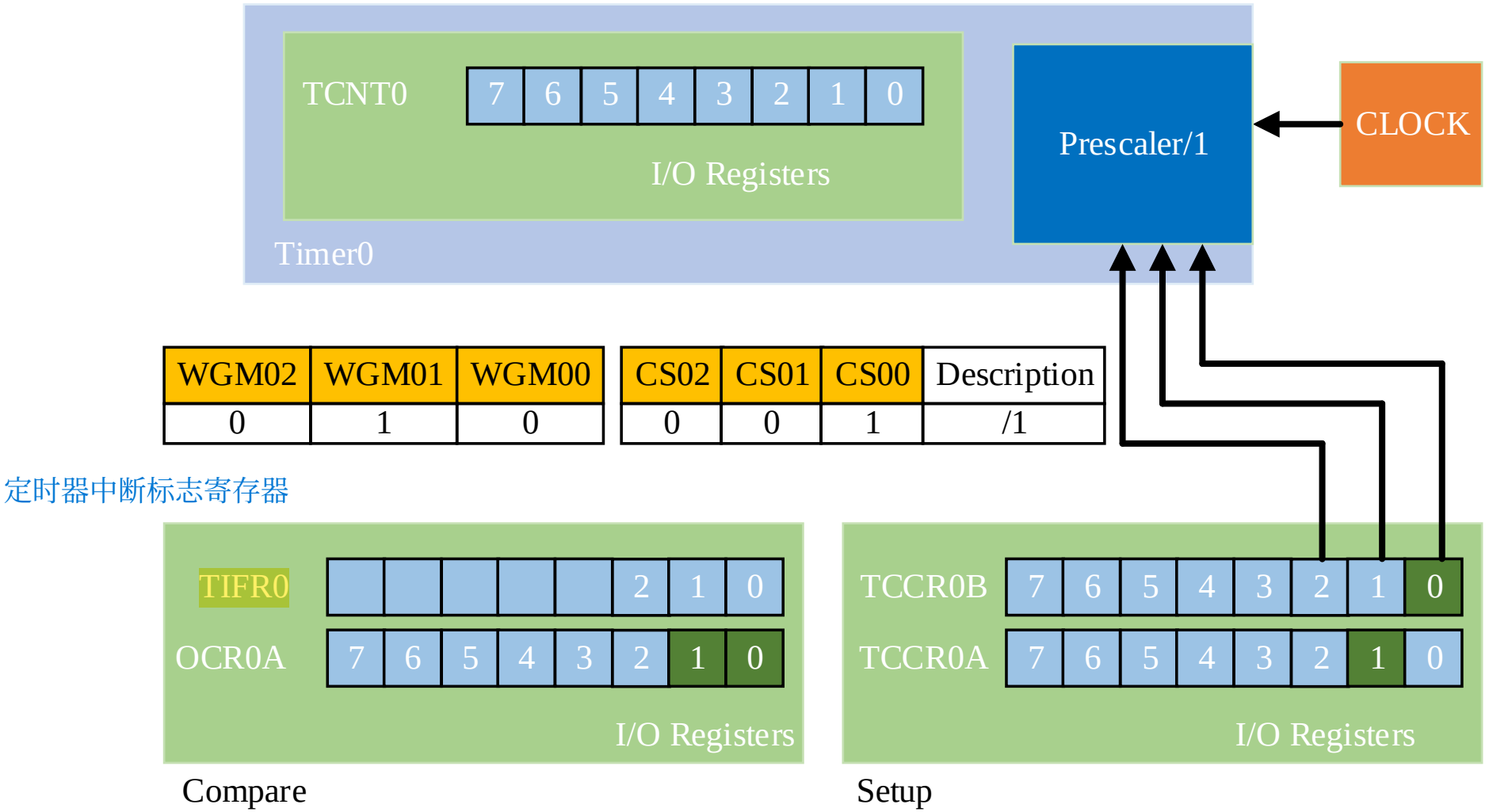
- Example where we set the MAX of the timer counter
  - Set OCR0A (Output Compare Register) to a value (used as the max of the counter)
  - Using TCCR0A (Timer Counter Control Register) set to 0b00000010

定时器计数器控制寄存器

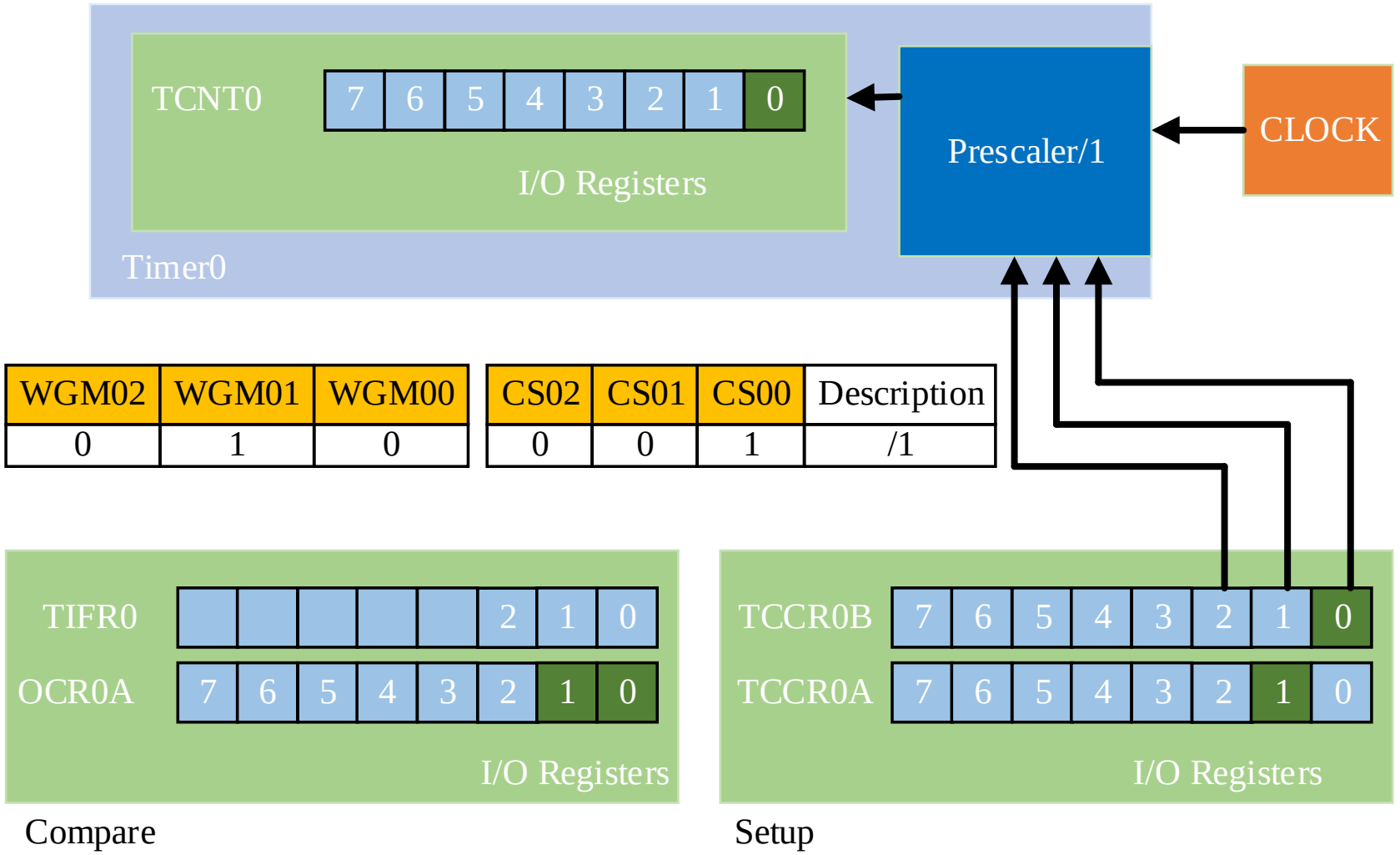
	7	6	5	4	3	2	1	0
TCCR0A	COM0A1	COM0A0	COM0B1	COM0B0	-	-	**WGM01**	**WGM00**
	7	6	5	4	3	2	1	0
TCCR0B	FOC0A	FOC0B	-	-	**WGM02**	CS02	CS01	CS00

I/O Registers

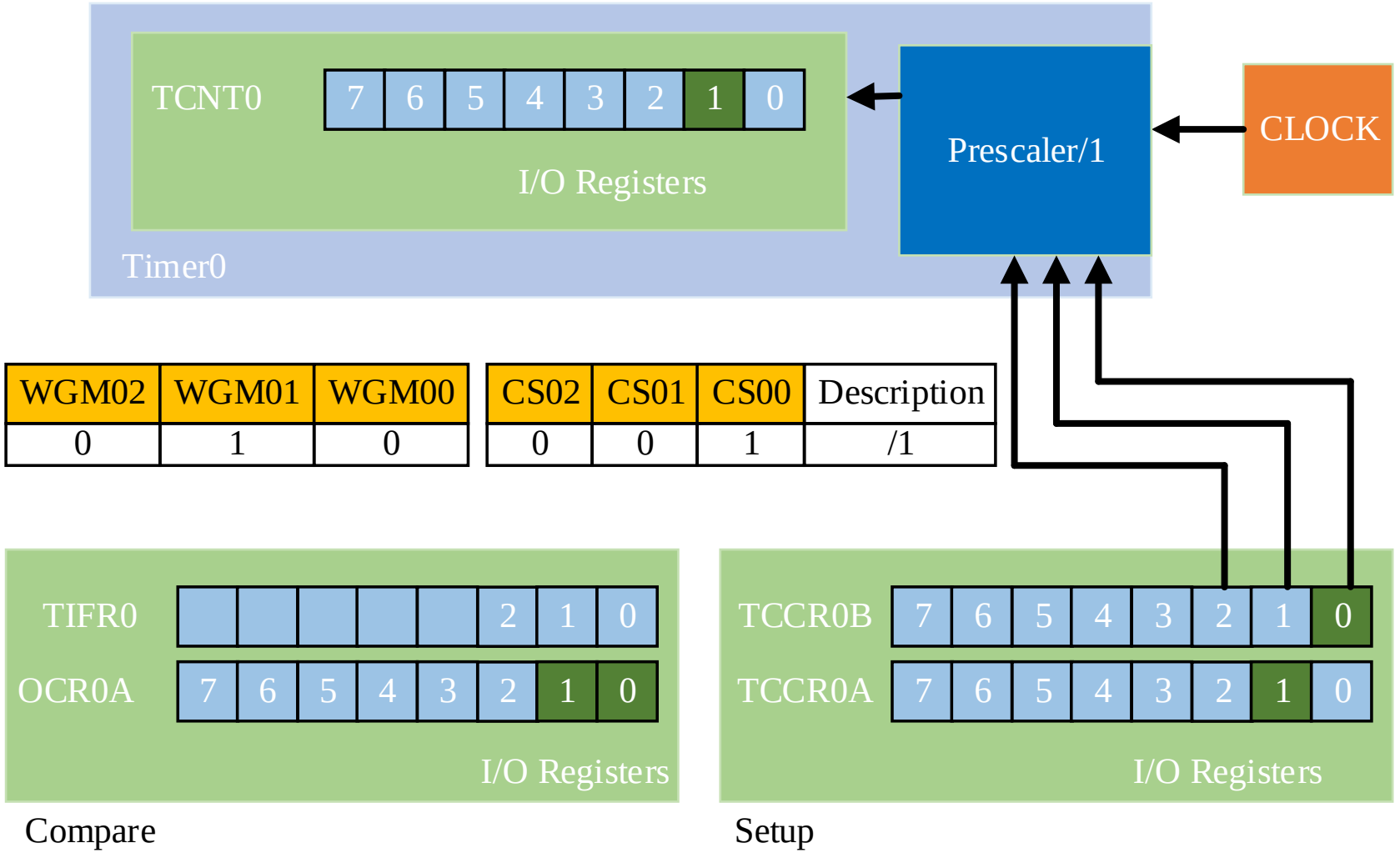
Set MAX - I


TCNT0
7 6 5 4 3 2 1 0
I/O Registers
Timer0
Prescaler/1
CLOCK
WGM02 WGM01 WGM00 CS02 CS01 CS00 Description
0 1 0 0 0 1 /1
定时器中断标志寄存器
TIFR0
2 1 0
OCR0A
7 6 5 4 3 2 1 0
I/O Registers
Compare
TCCR0B
7 6 5 4 3 2 1 0
TCCR0A
7 6 5 4 3 2 1 0
I/O Registers
Setup

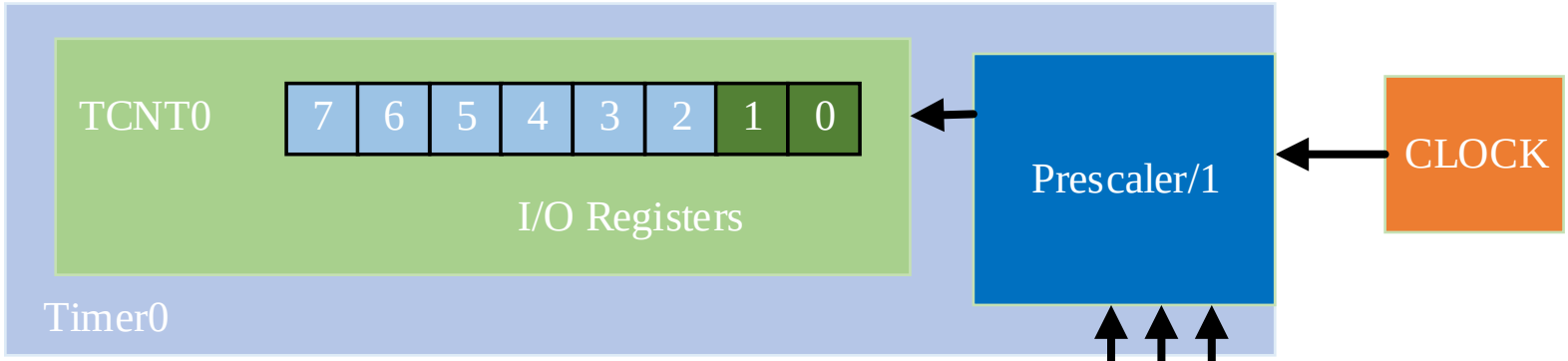
Set MAX - II


TCNT0
7 6 5 4 3 2 1 0
I/O Registers
Timer0
Prescaler/1
CLOCK
WGM02 WGM01 WGM00
0 1 0
CS02 CS01 CS00 Description
0 0 1 /1
TIFR0
2 1 0
OCR0A
7 6 5 4 3 2 1 0
I/O Registers
Compare
TCCR0B
7 6 5 4 3 2 1 0
TCCR0A
7 6 5 4 3 2 1 0
I/O Registers
Setup

Set MAX - III

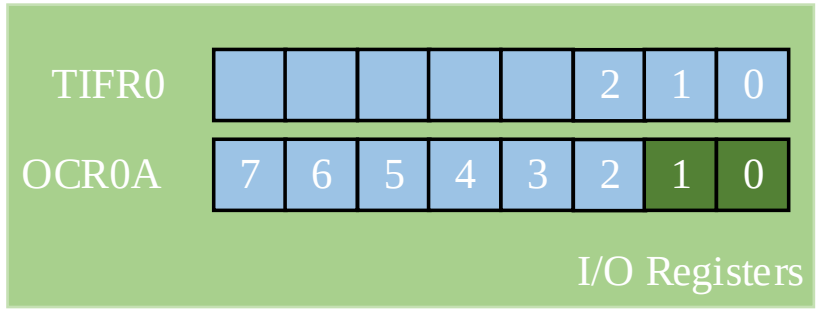

TCNT0
7 6 5 4 3 2 1 0
I/O Registers
Timer0
Prescaler/1
CLOCK
WGM02 WGM01 WGM00
0 1 0
CS02 CS01 CS00 Description
0 0 1 /1
TIFR0
2 1 0
OCR0A
7 6 5 4 3 2 1 0
I/O Registers
Compare
TCCR0B
7 6 5 4 3 2 1 0
TCCR0A
7 6 5 4 3 2 1 0
I/O Registers
Setup

Set MAX - IV

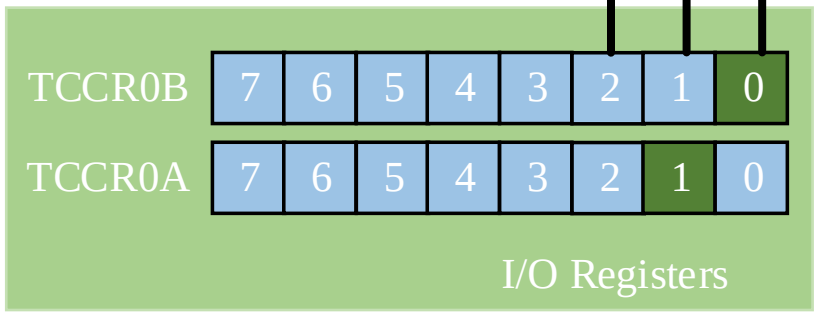


WGM02	WGM01	WGM00
0	1	0

CS02	CS01	CS00	Description
0	0	1	/1

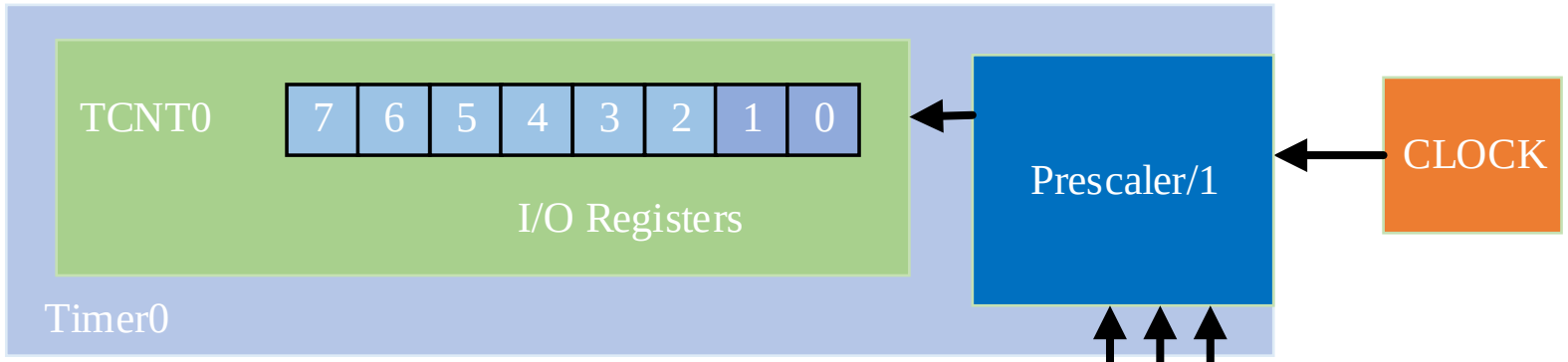


Compare



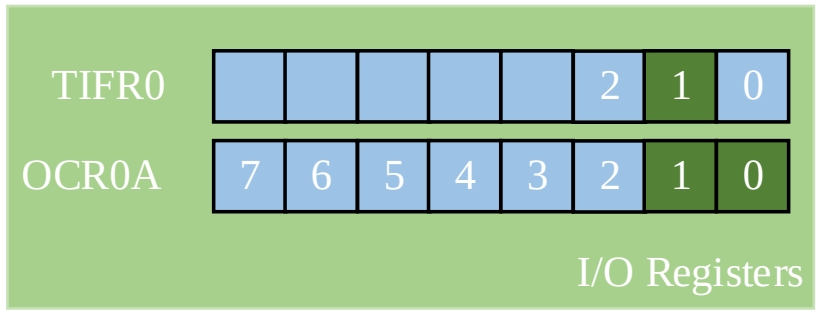
Setup

Set MAX - V

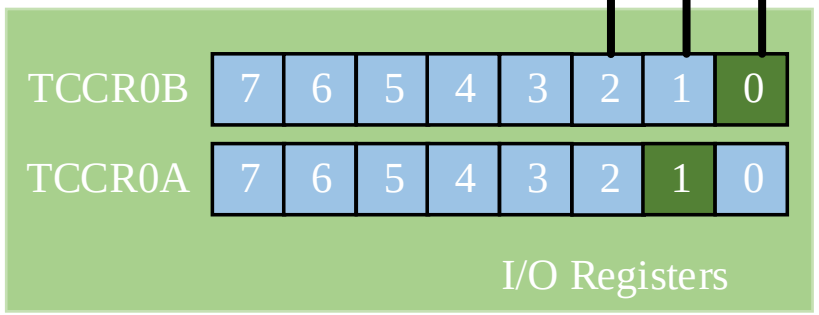


WGM02	WGM01	WGM00
0	1	0

CS02	CS01	CS00	Description
0	0	1	/1



Compare

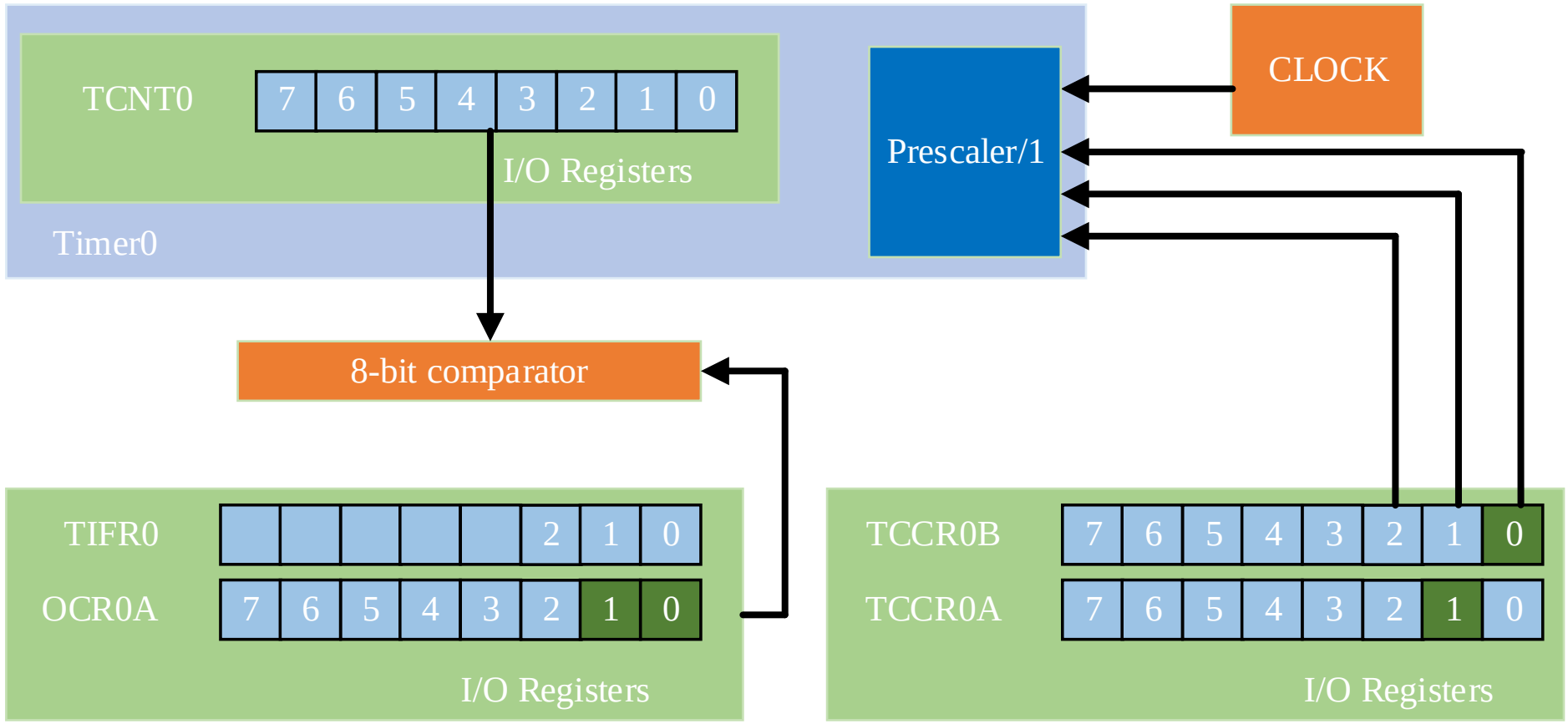


Setup

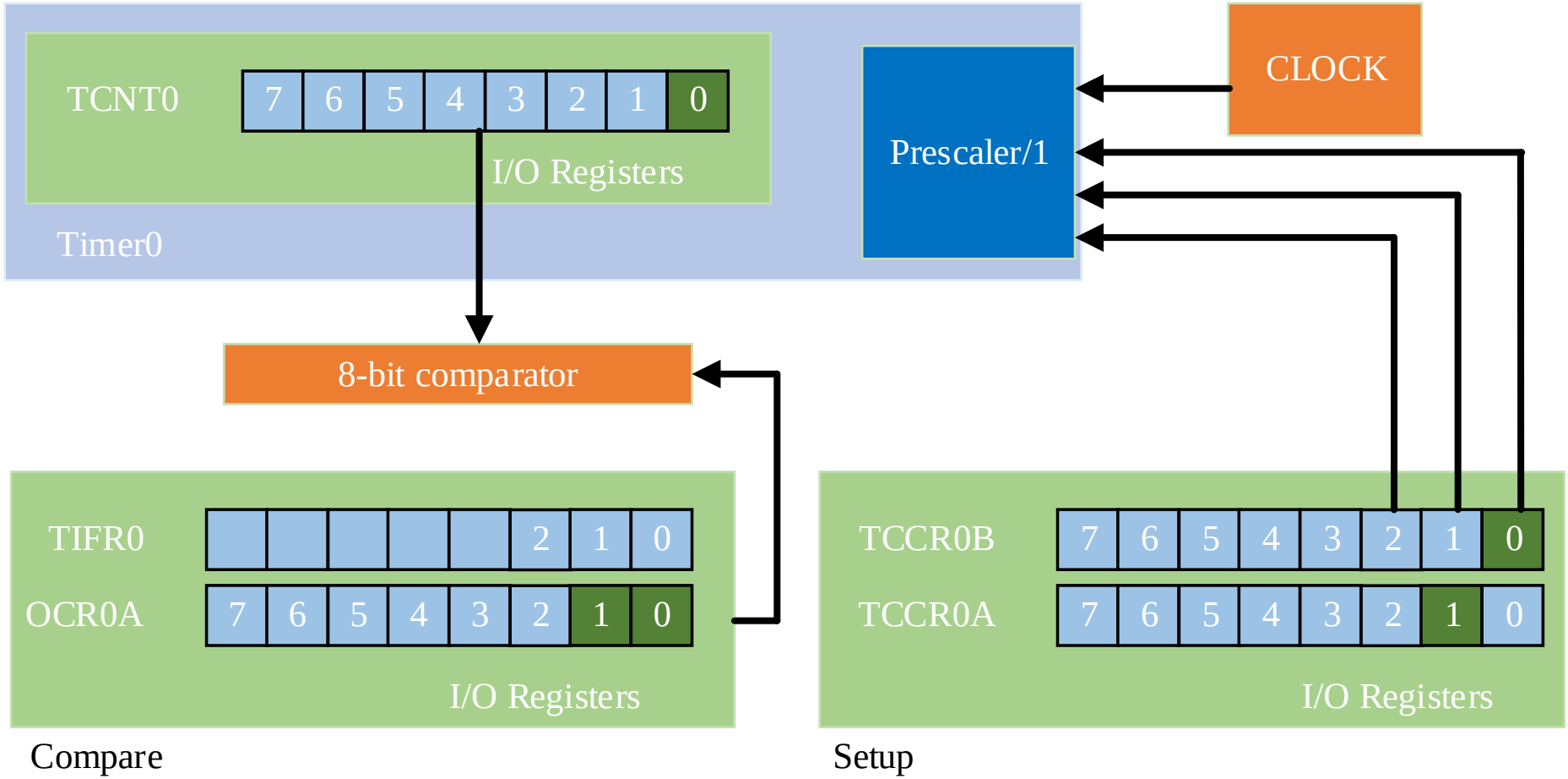
What is really happening?

- How does the microntroller knows the current value has reached the new MAX?

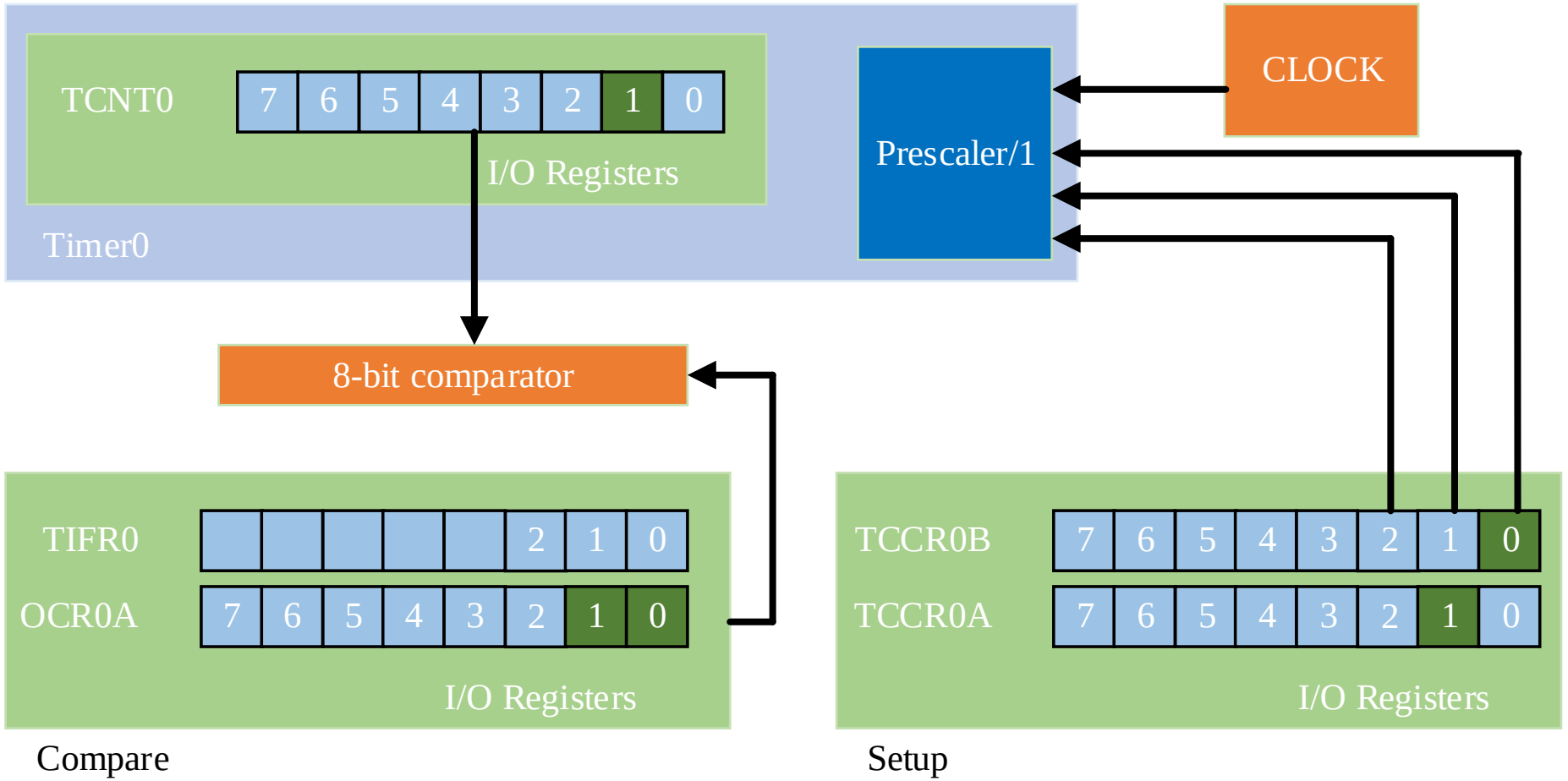
Output Comparator - I



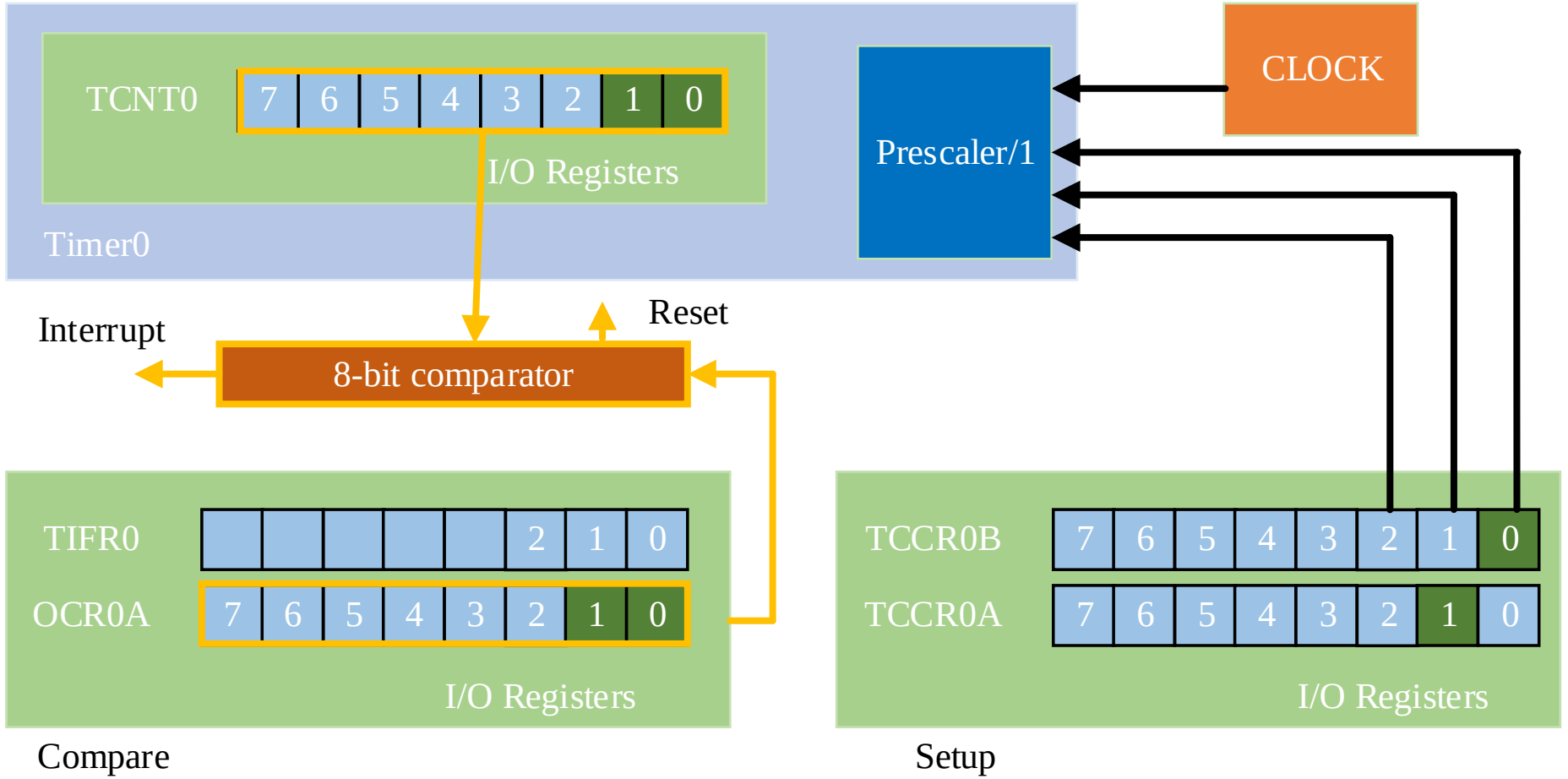
Output Comparator - II



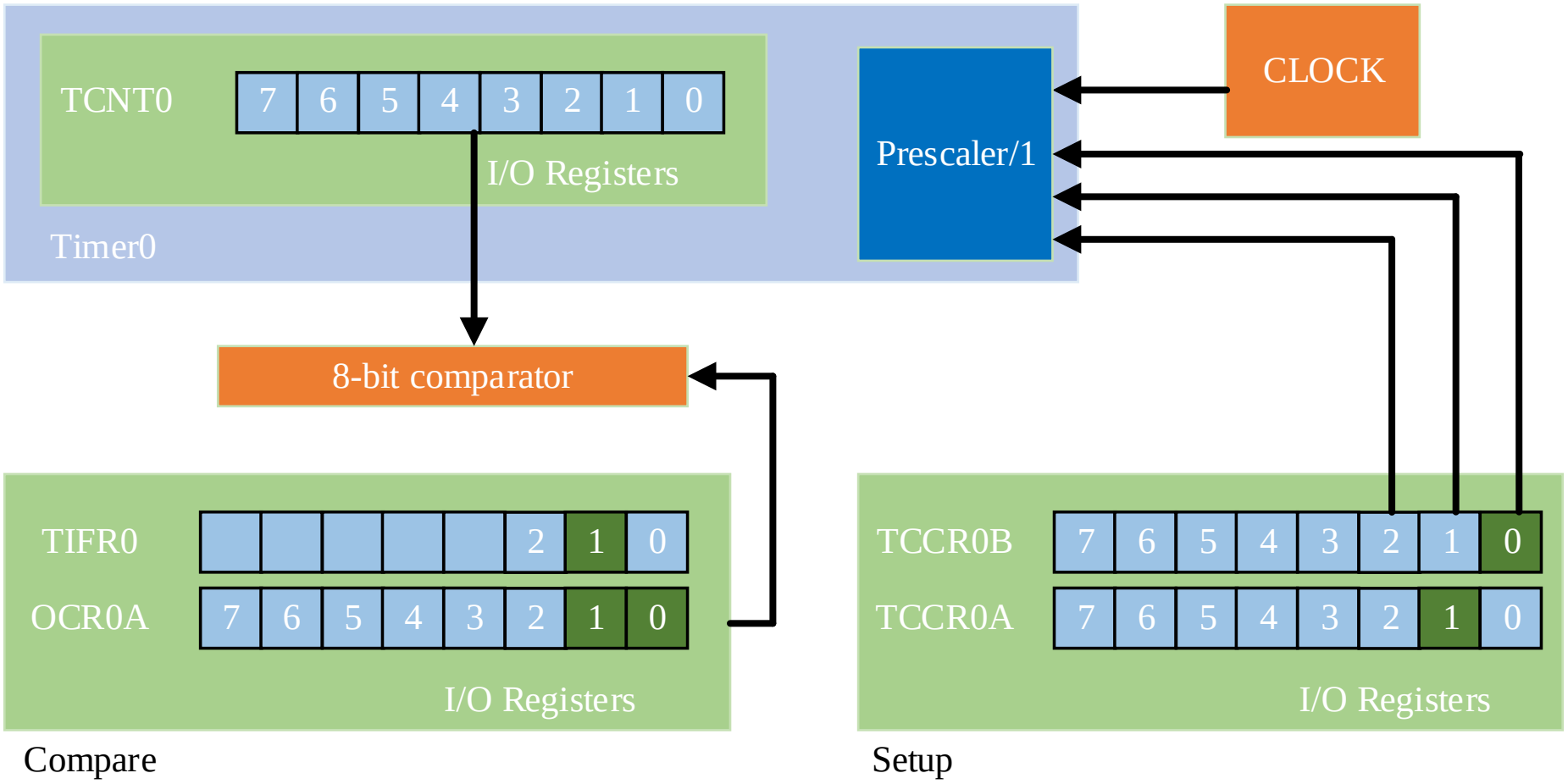
Output Comparator - III



Output Comparator - IV



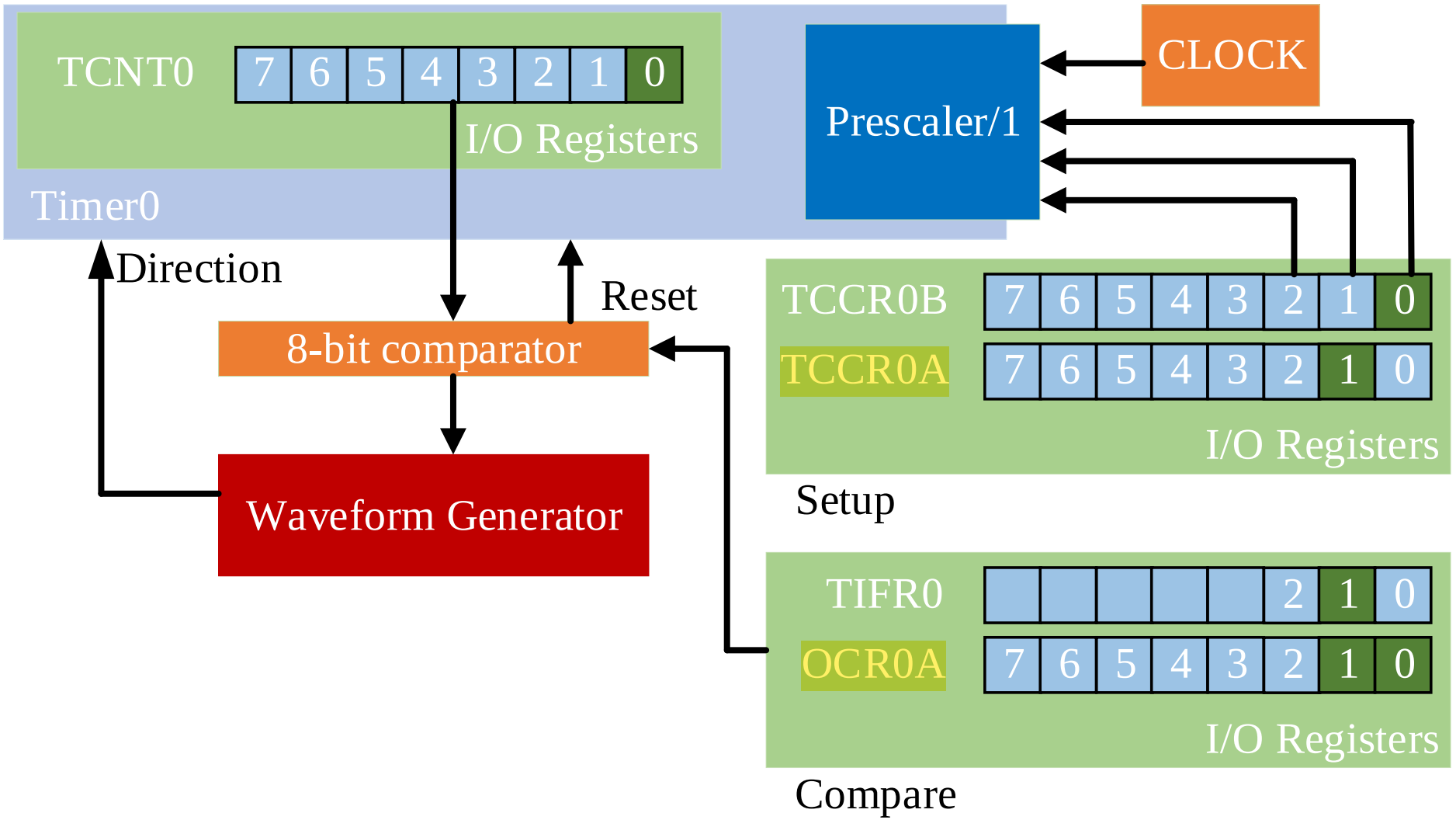
Output Comparator - V



So what?

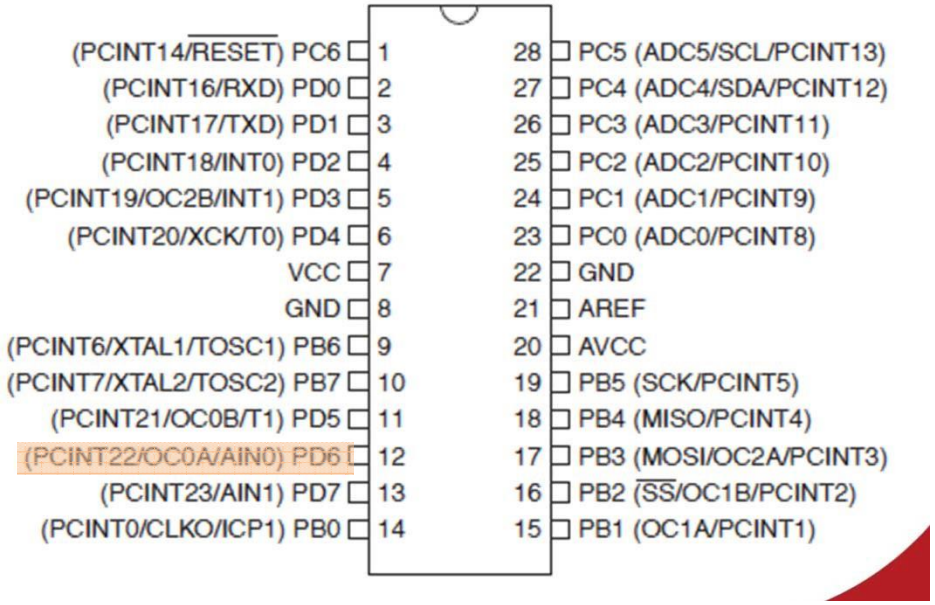
- We can use the comparator to generate waveforms.

Output Compare

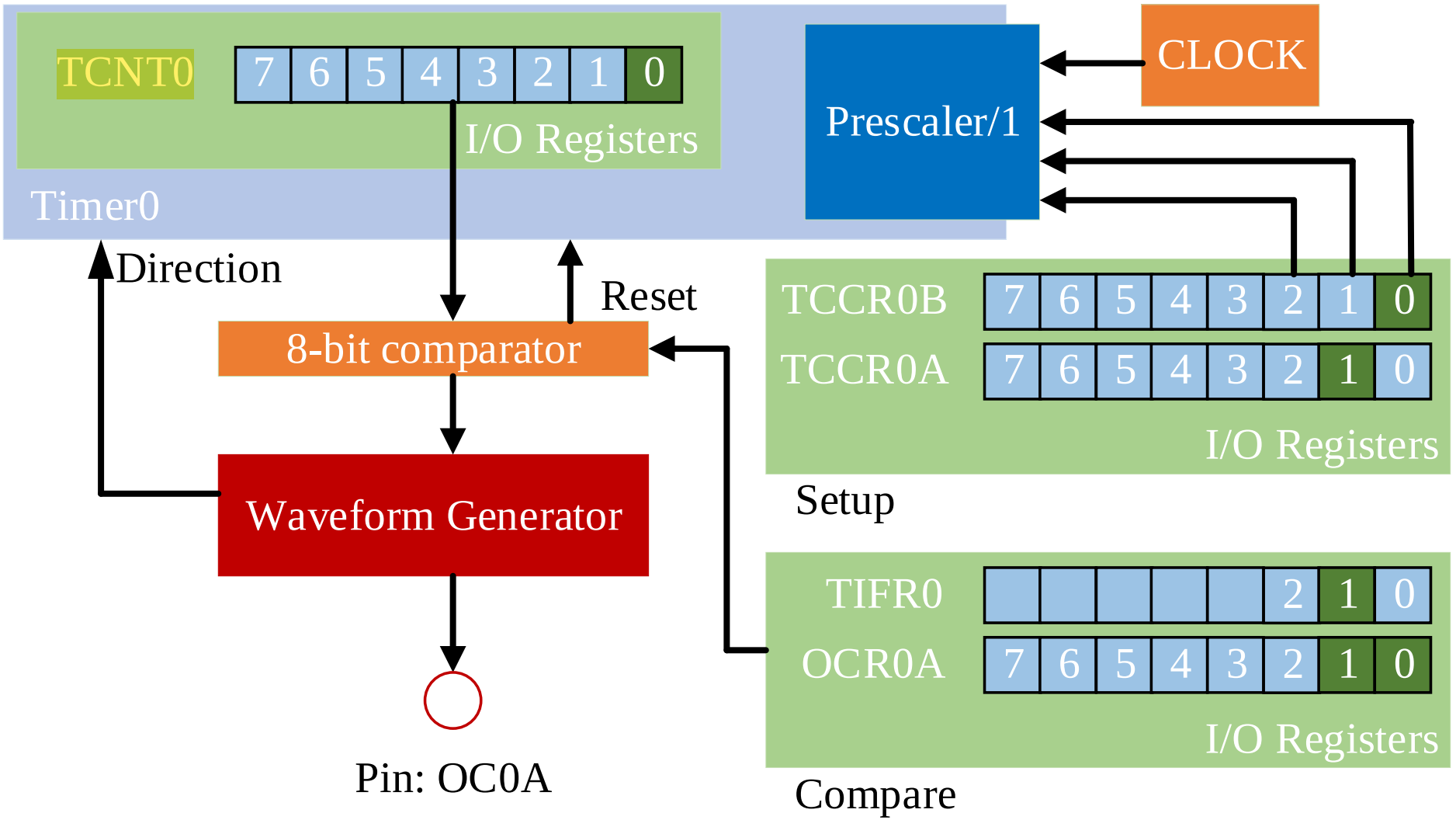


Waveform generator

- A waveform generator is connected directly to an output pin
  - In this case OC0A (PortD6)
  - Overrides value in register PortD
  - **Only works when pin is set to output!**



Output Compare



Waveform generator

- Can operate in a number of modes
- Controlled by 3 bits in two control registers
- 3 bits WGM0n set the mode
- Used before (when setting the max of the timer)

定时器/计数器控制寄存器

	7	6	5	4	3	2	1	0
TCCR0A	COM0A1	COM0A0	COM0B1	COM0B0	-	-	WGM01	WGM00
	7	6	5	4	3	2	1	0
TCCR0B	FOC0A	FOC0B	-	-	WGM02	CS02	CS01	CS00

I/O Registers

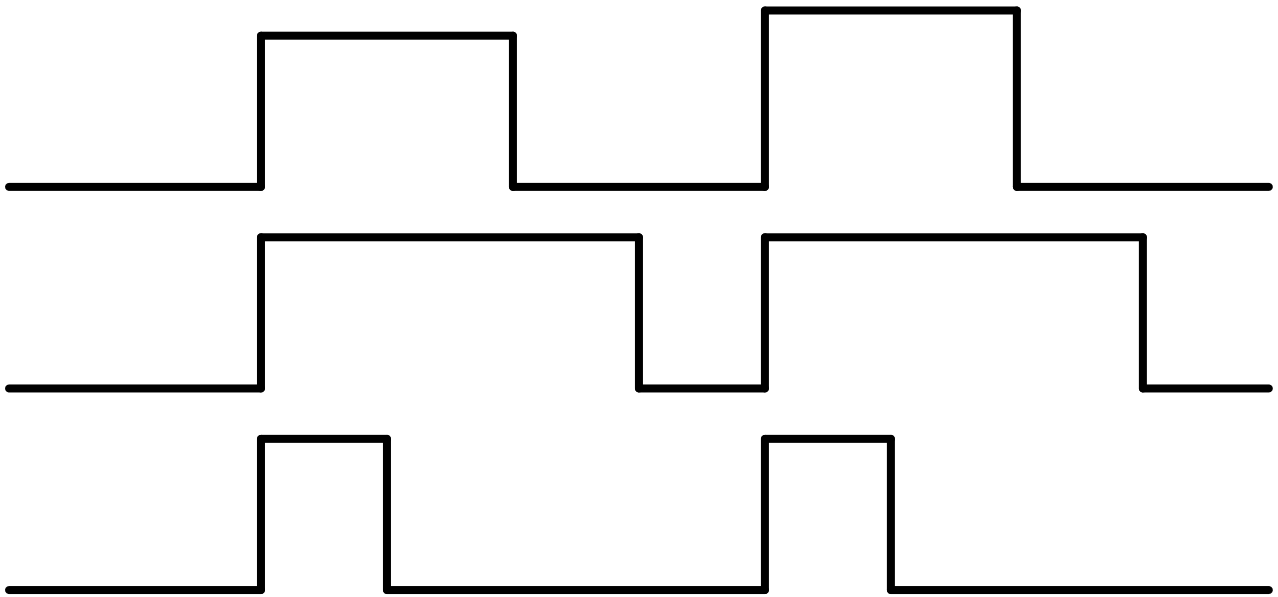
Choose your mode

Waveform Generation Mode Bit Description

Mode	WGM 02	WGM 01	WGM 00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on
0	0	0	0	Normal	0xFF	immediate	MAX
1	0	0	1	PWM, Phase correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	-	-	-
5	1	0	1	PWM, Phase correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	-	-	-
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

Used highlighted mode already to clear the timer on OCRA

Frequency control

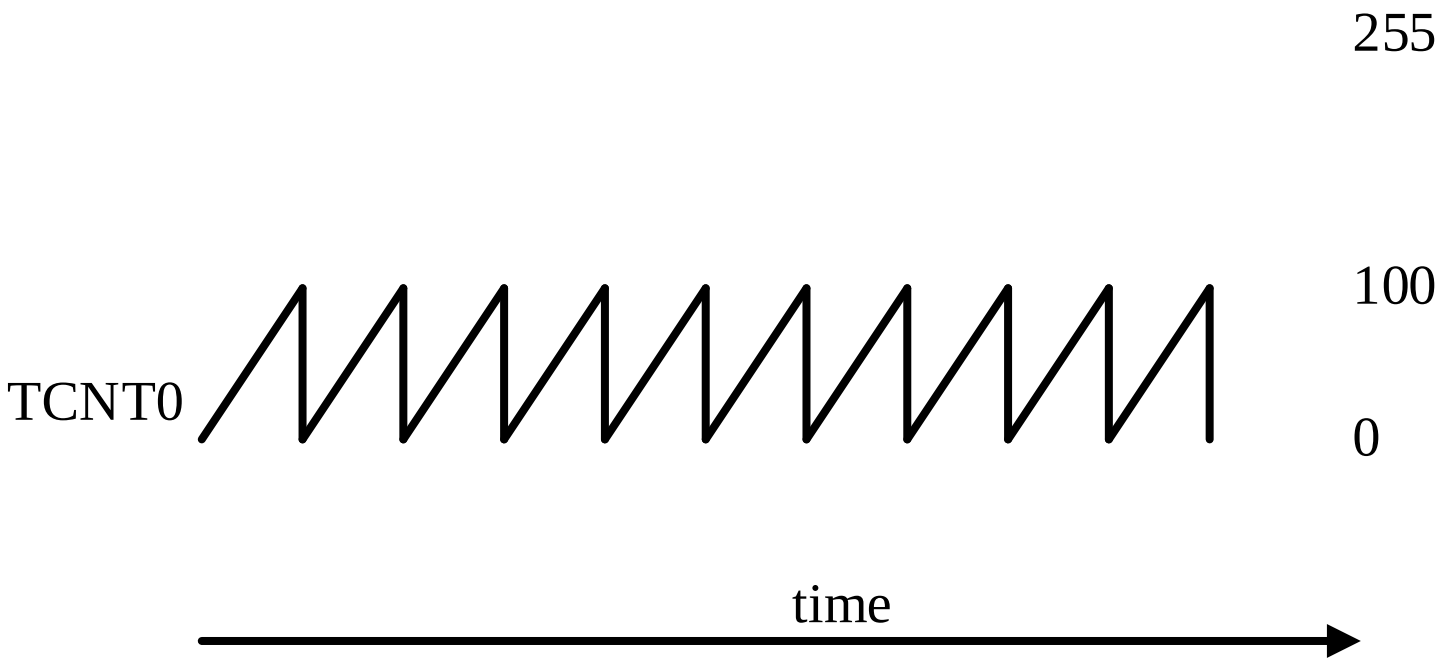


50% Duty Cycle

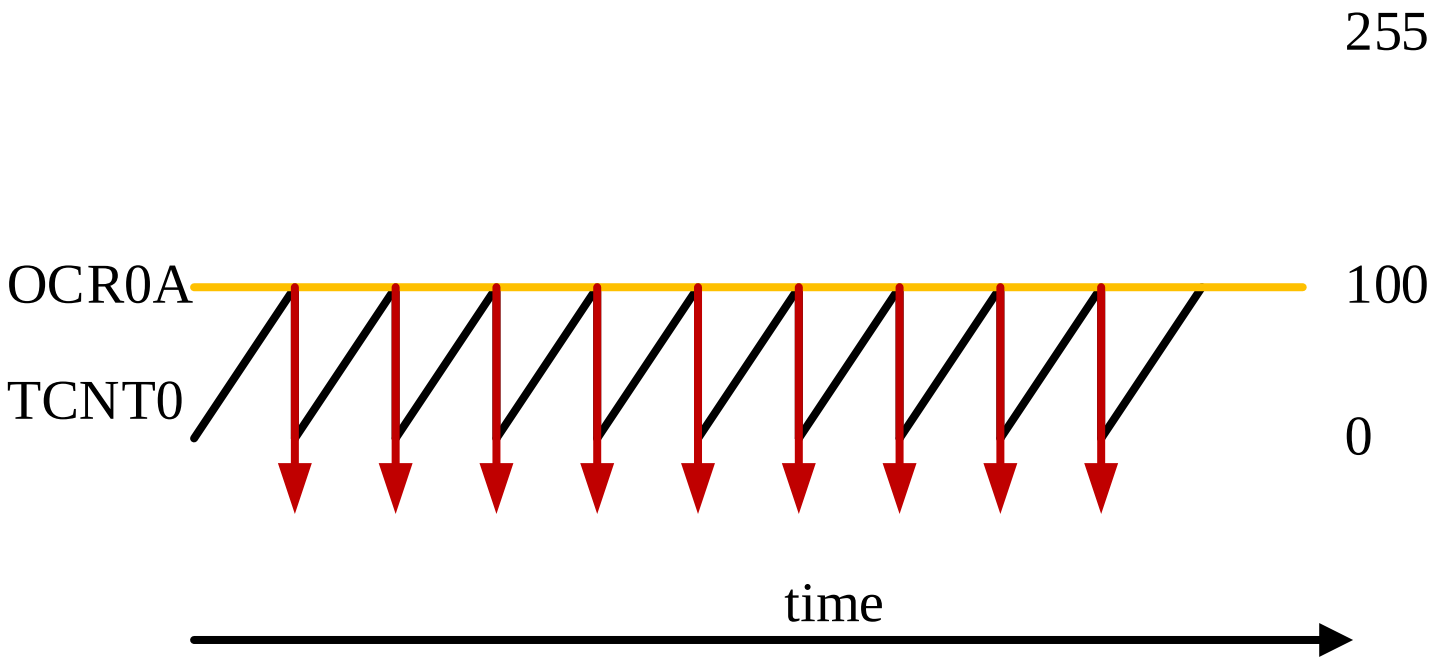
75% Duty Cycle

25% Duty Cycle

Timing Diagram - I

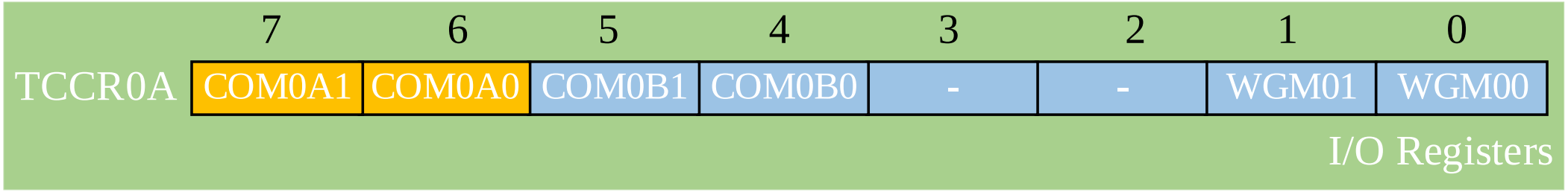


Timing Diagram - II



Timer Counter Control Register A

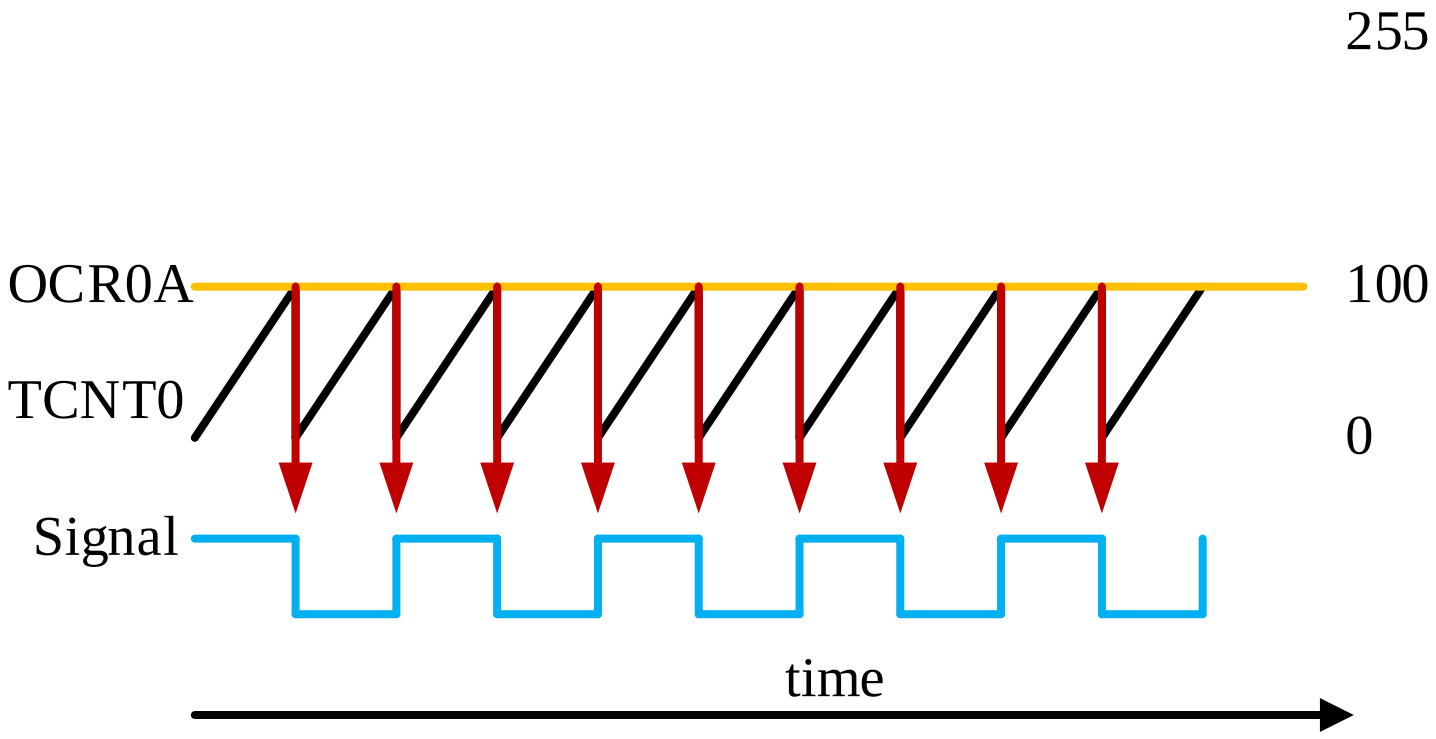
- Choose behaviour of this output pin.
  - Highlighted setting is normally best (toggling)



Compare Output Mode, non-PWM Mode

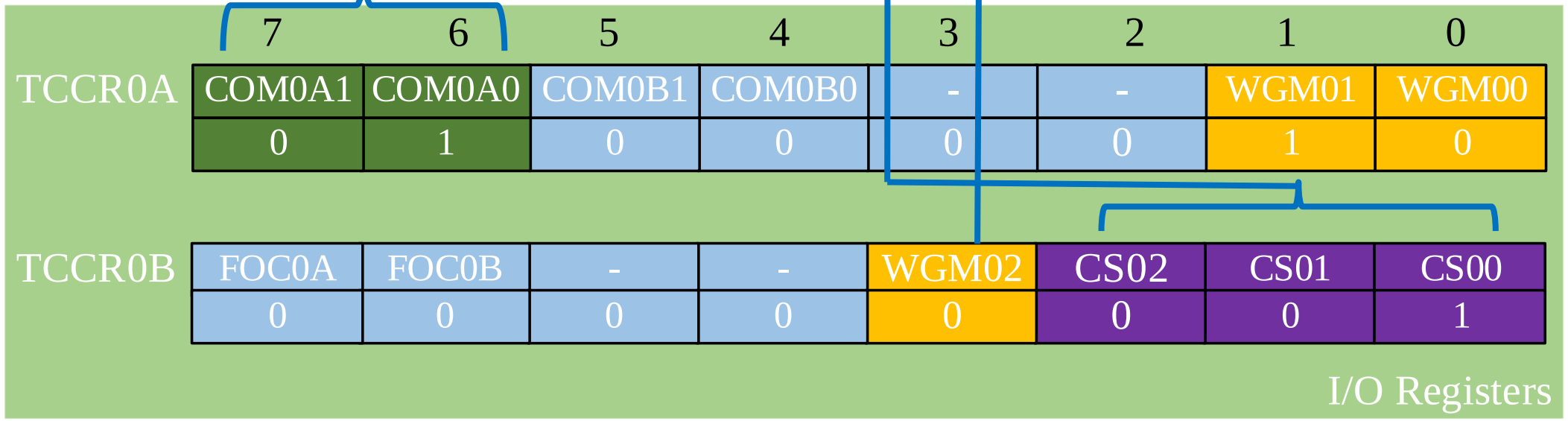
COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected
0	1	Toggle OC0A on Compare Match
1	0	Clear OC0A on Compare Match
1	1	Set OC0A on Compare Match

Timing Diagram

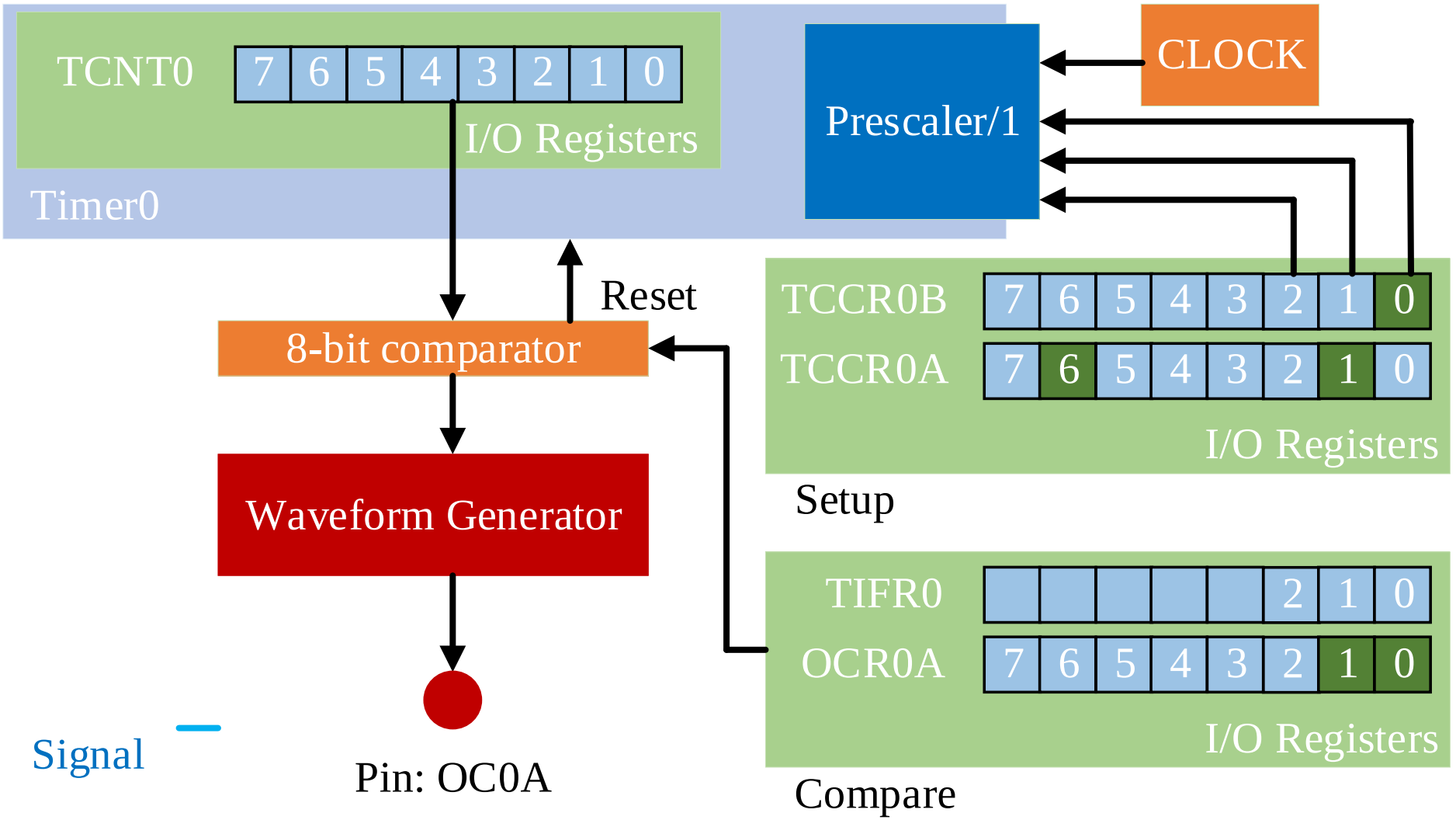


Clear timer on compare match

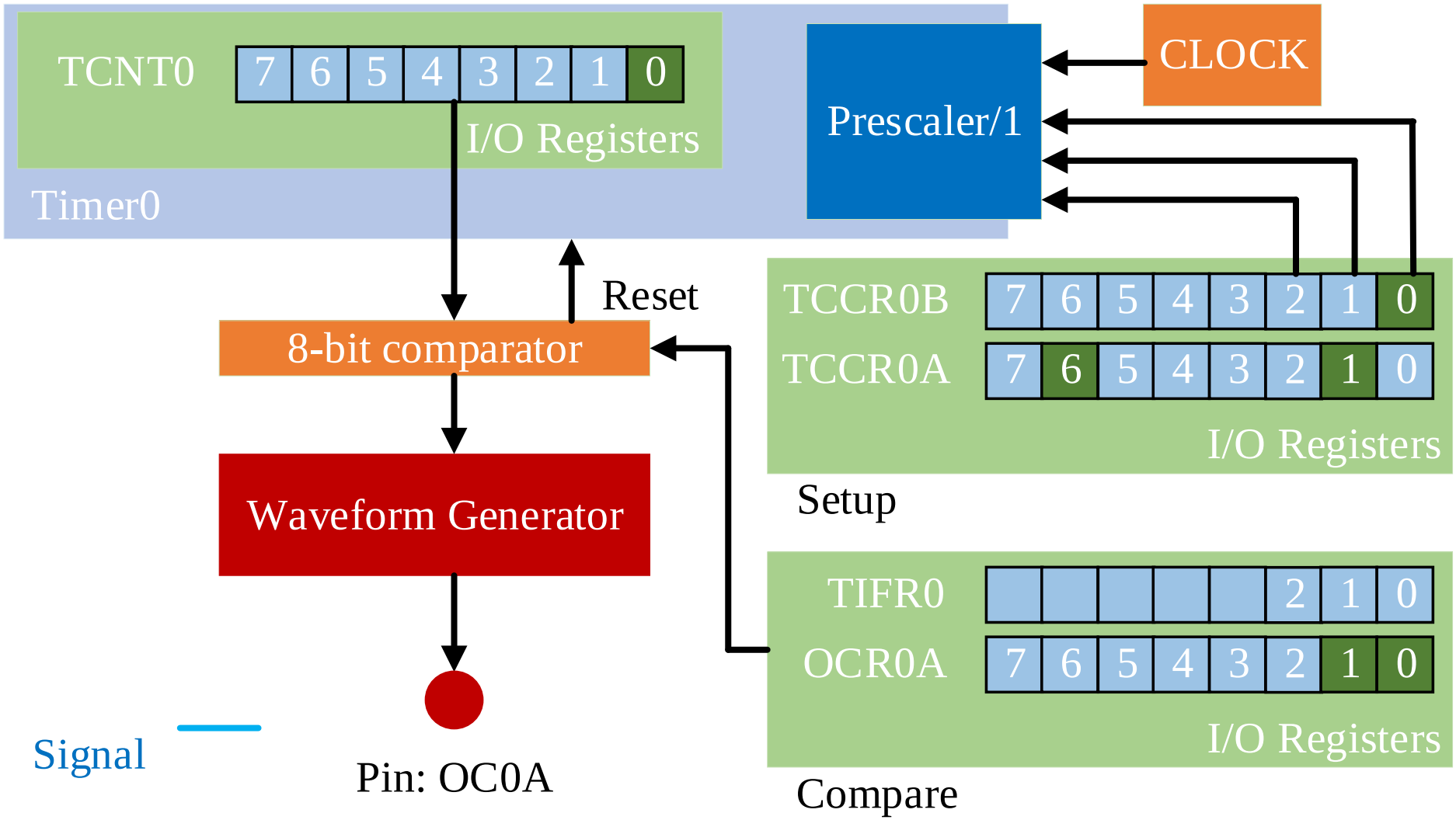
- A very simple timer mode
- Resets the counter when it reaches the value of OCR0A
- Same operation we have just reviewed
- WGM0n = 010 (CTC mode) [new max!]
- COM0An = 01 (Toggle pin on match)
- CS02 = 001 (/1)



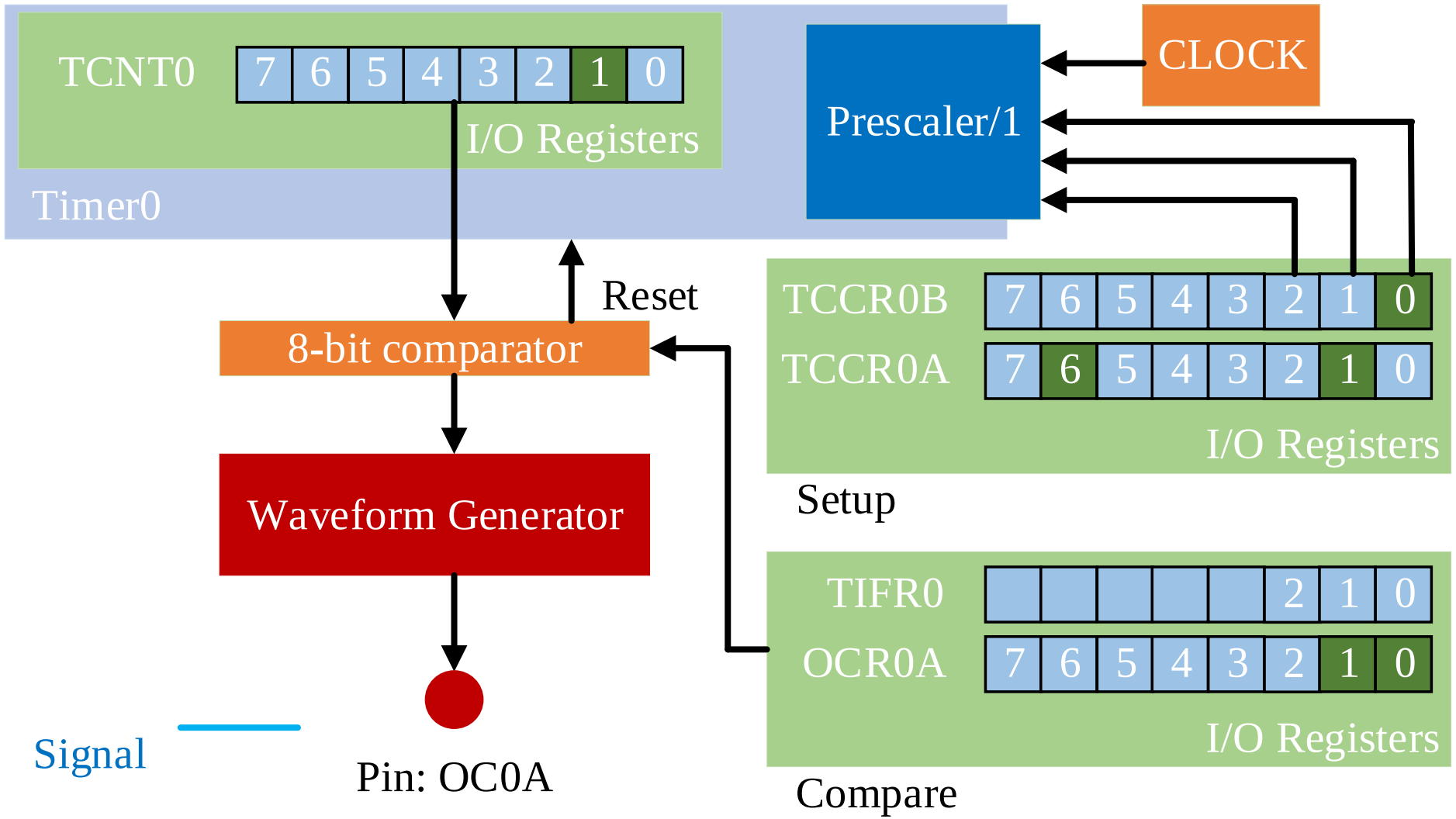
Frequency Control - I



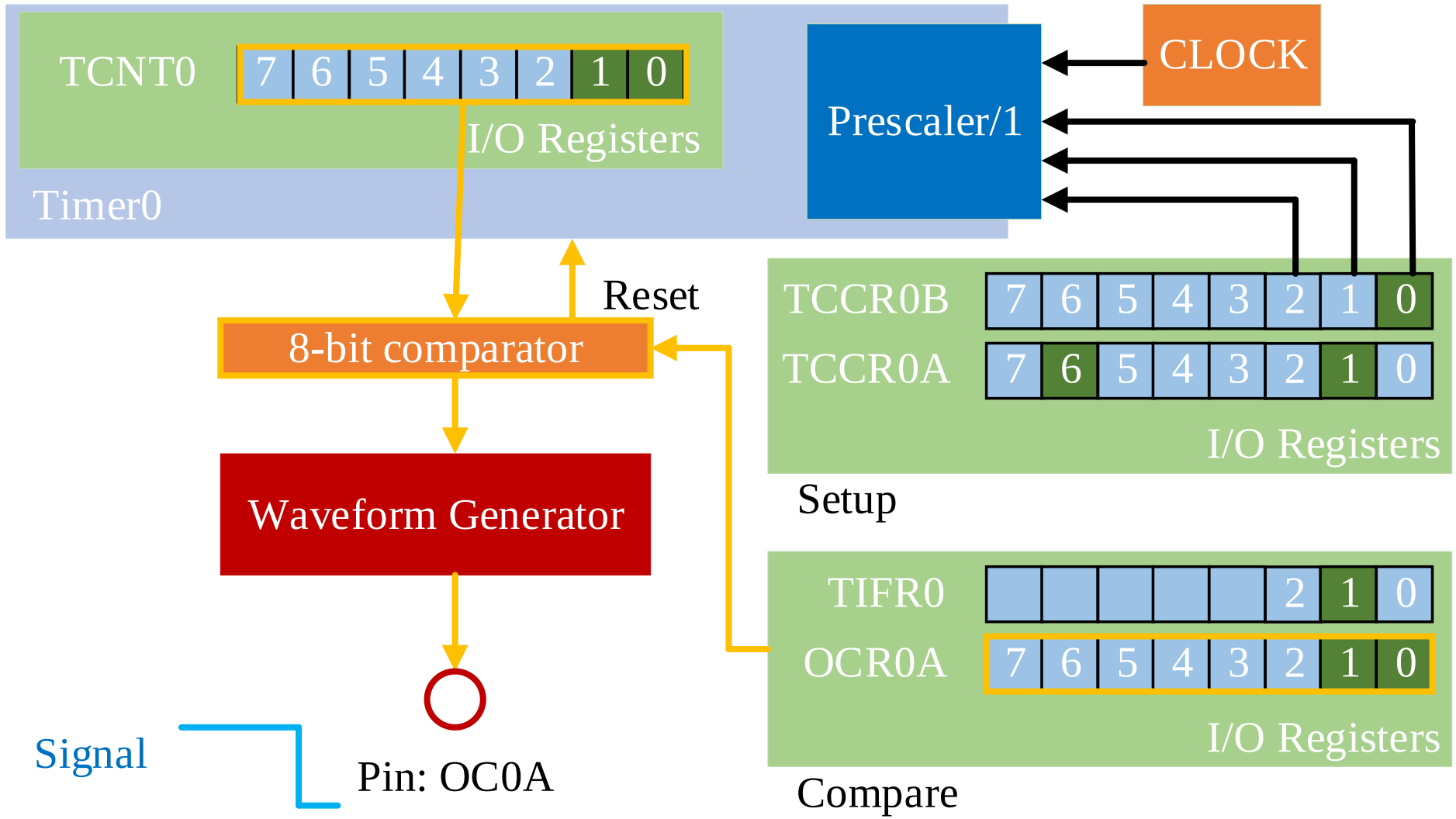
Frequency Control - II



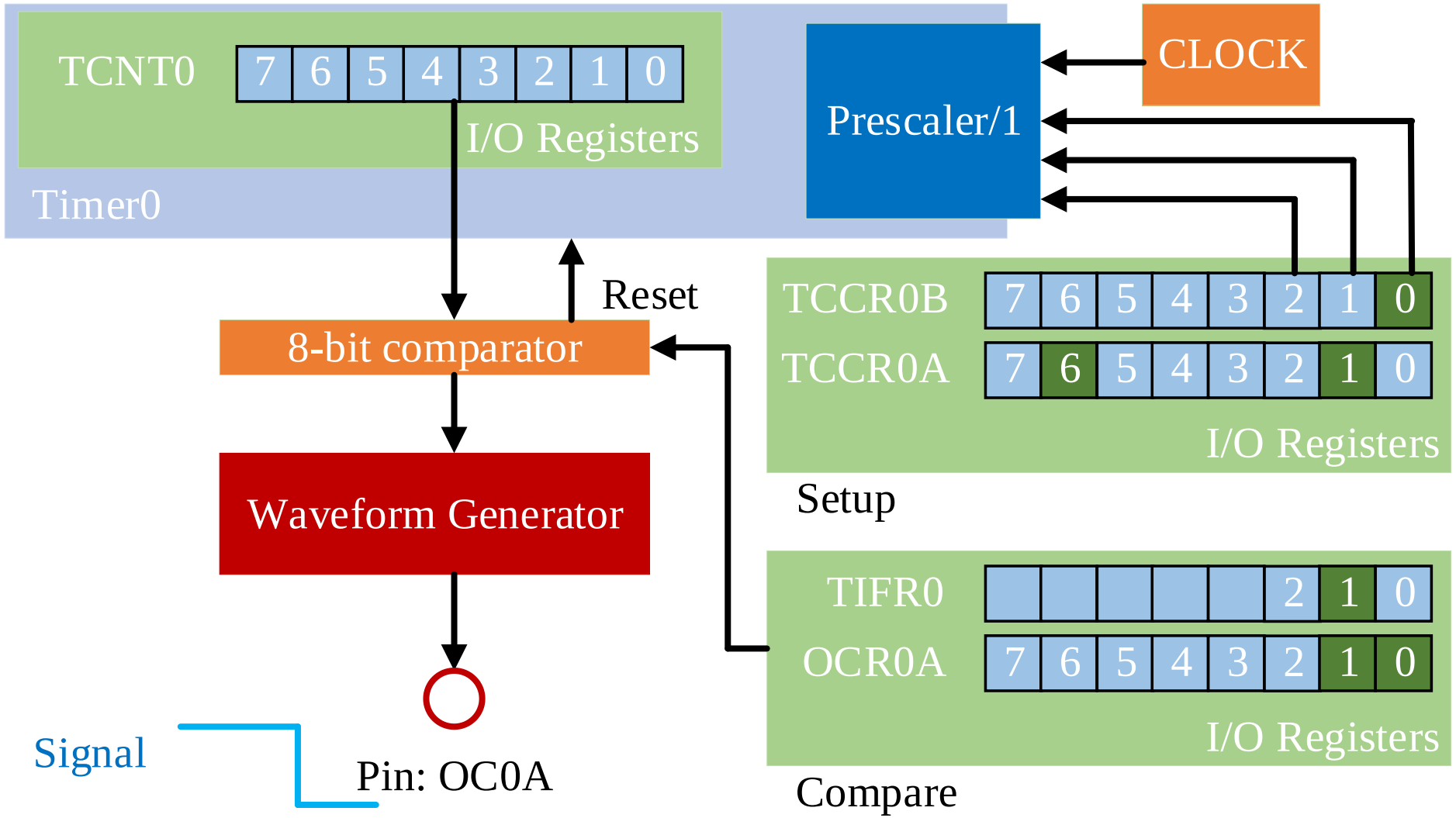
Frequency Control - III



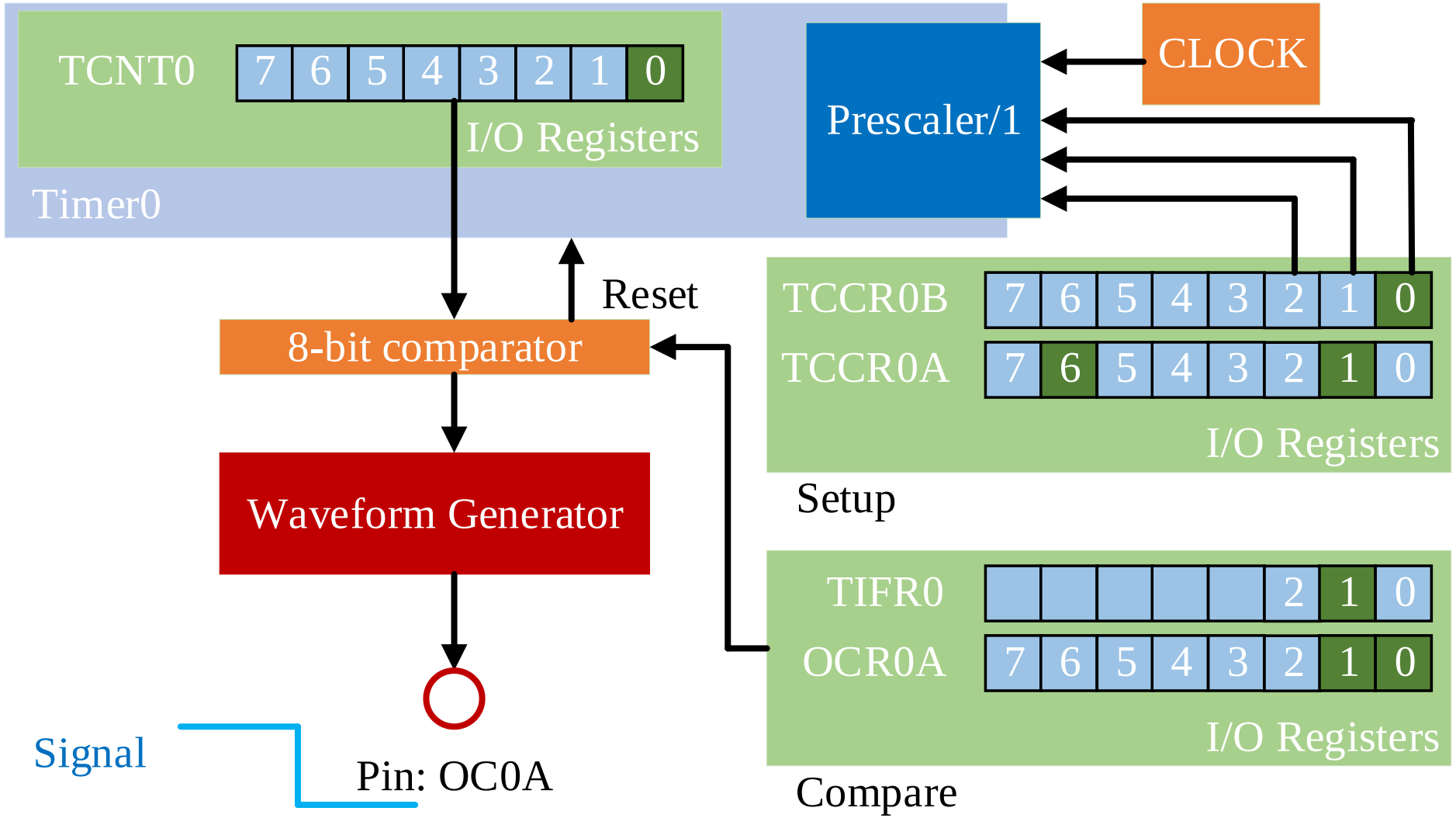
Frequency Control - IV



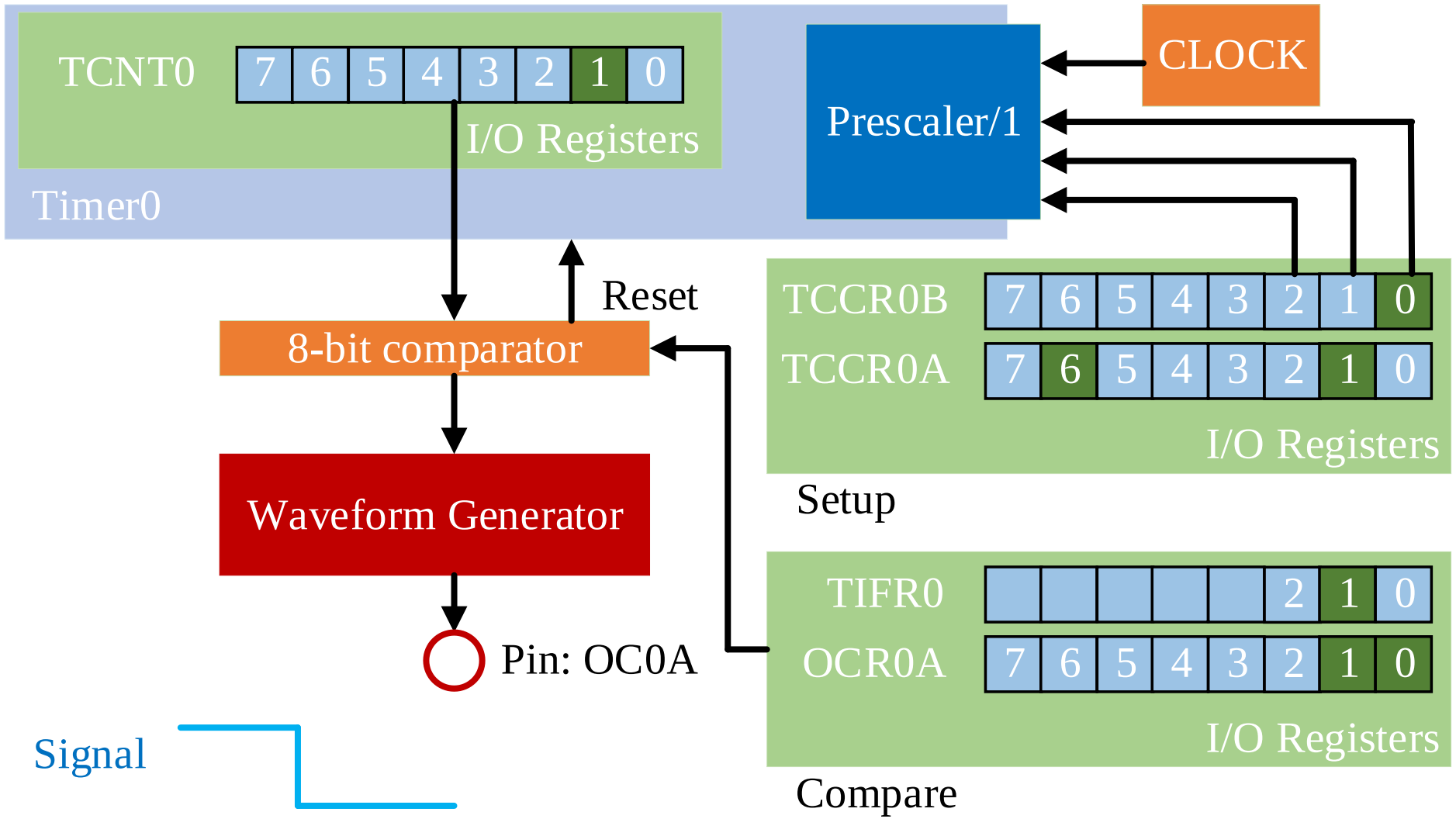
Frequency Control - V



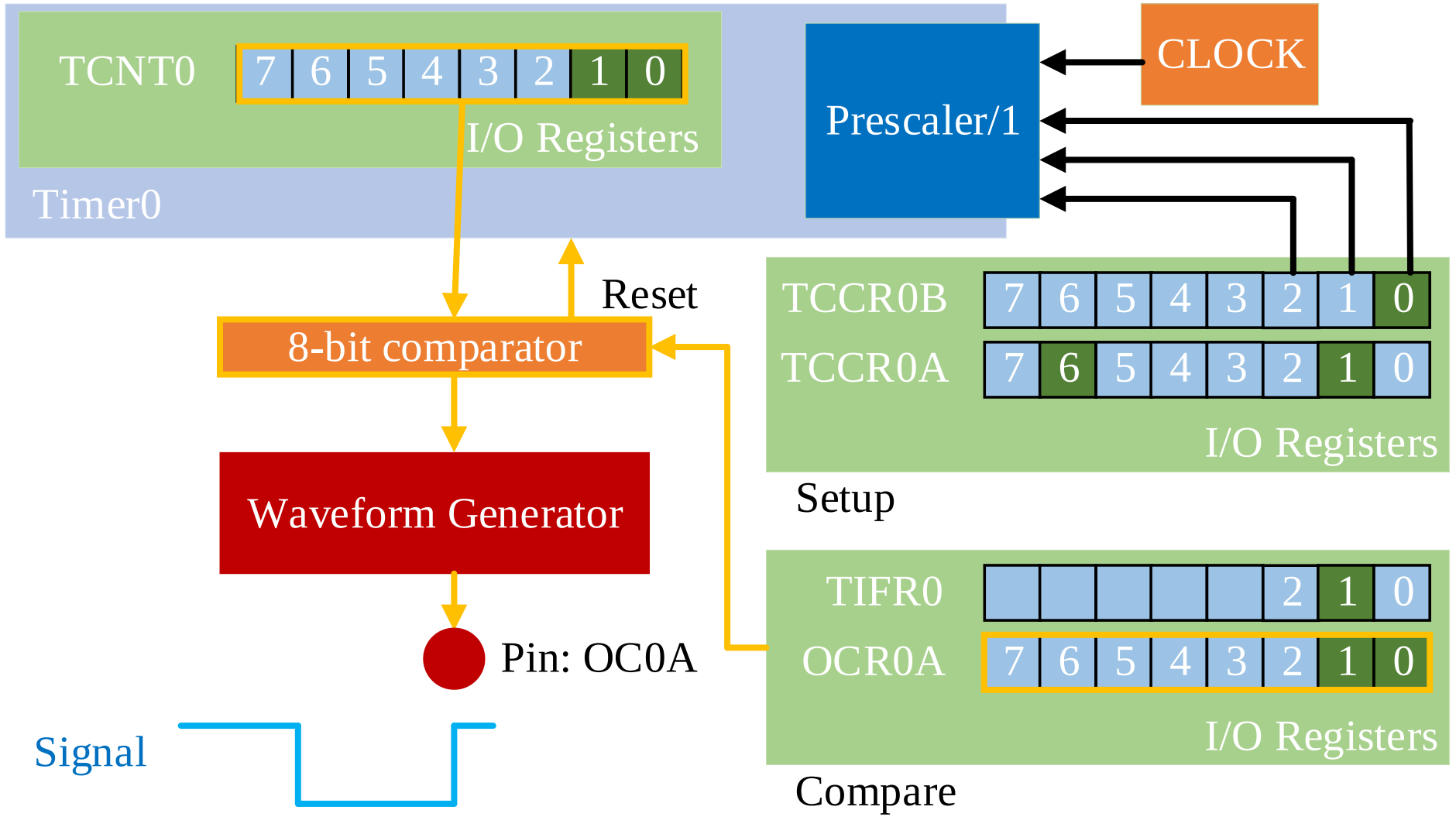
Frequency Control - VI



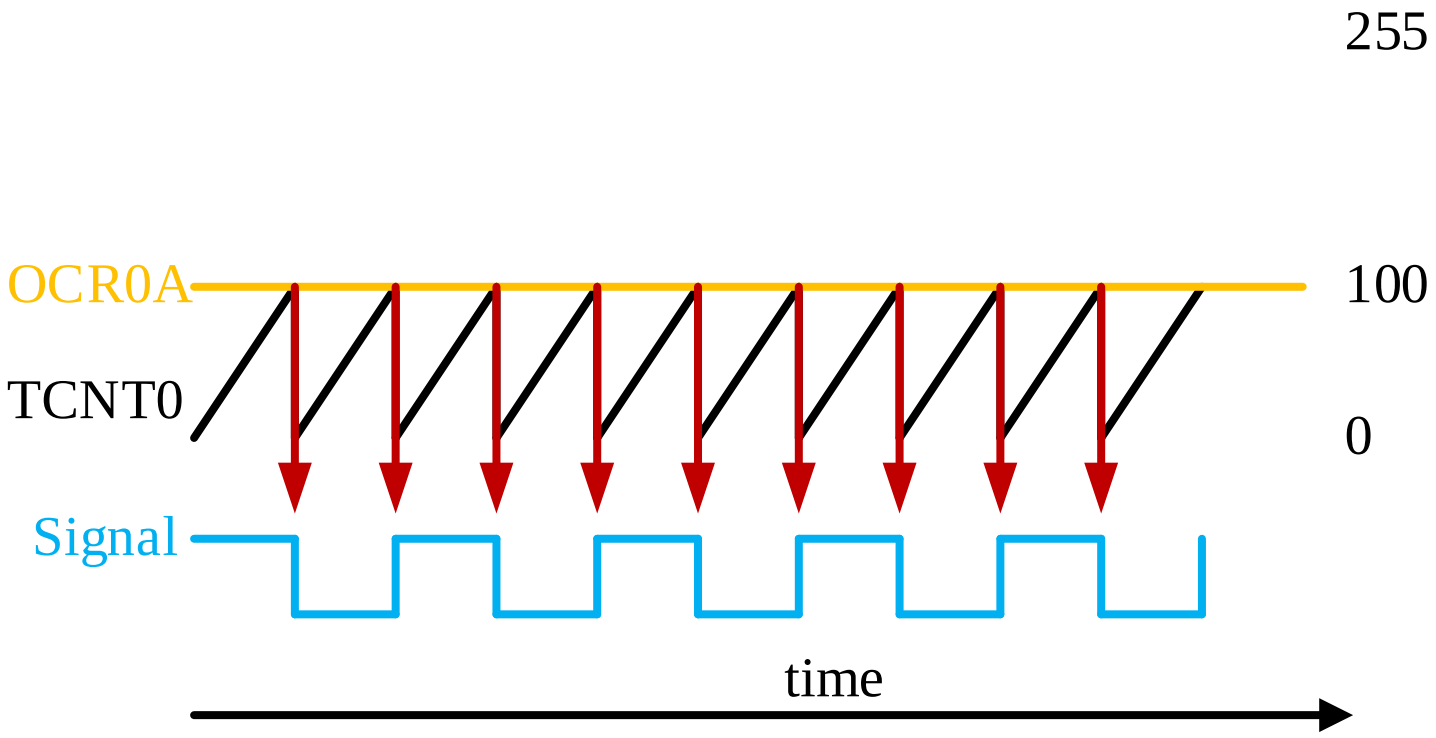
Frequency Control - VII



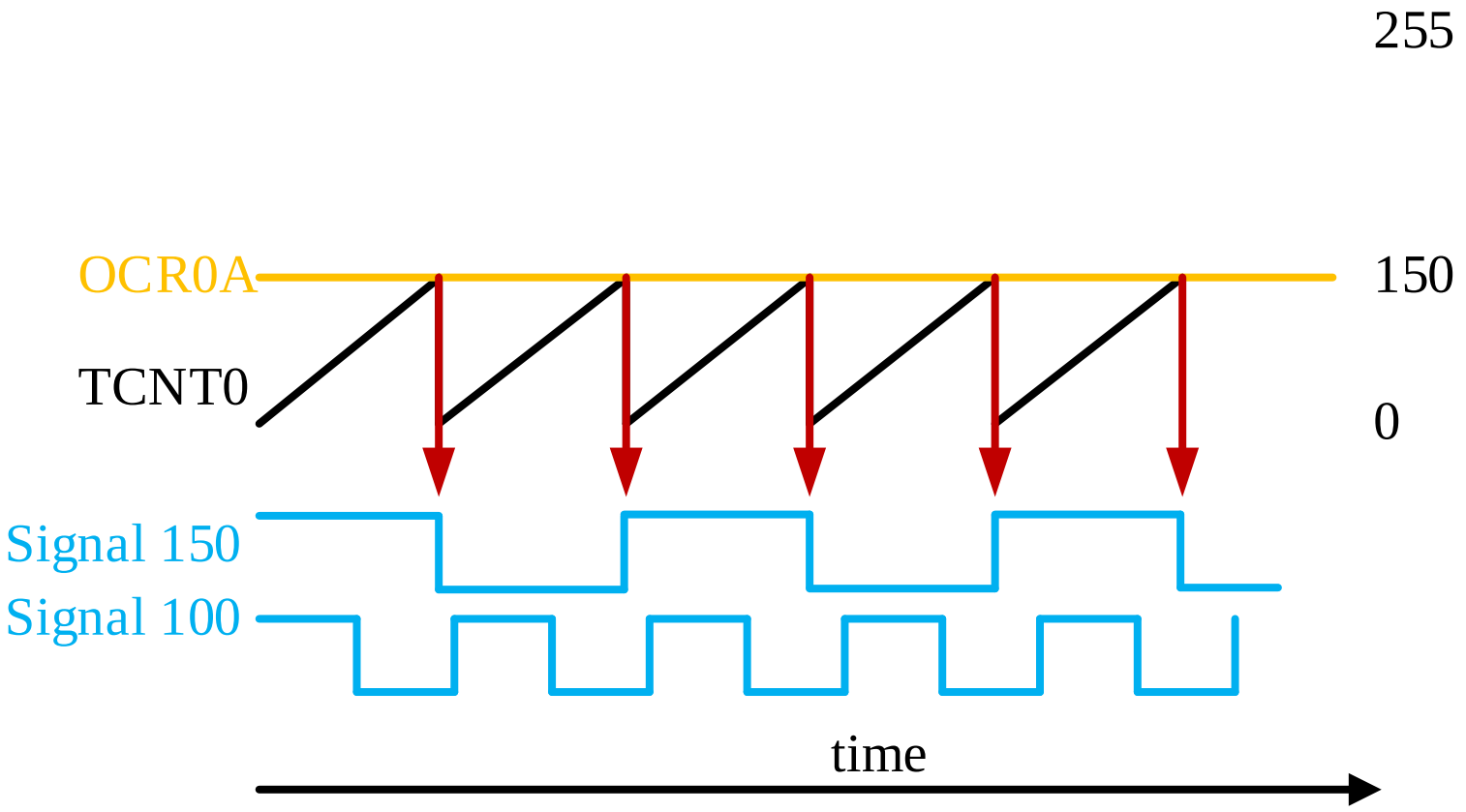
Frequency Control - VIII



Timing Diagram - I



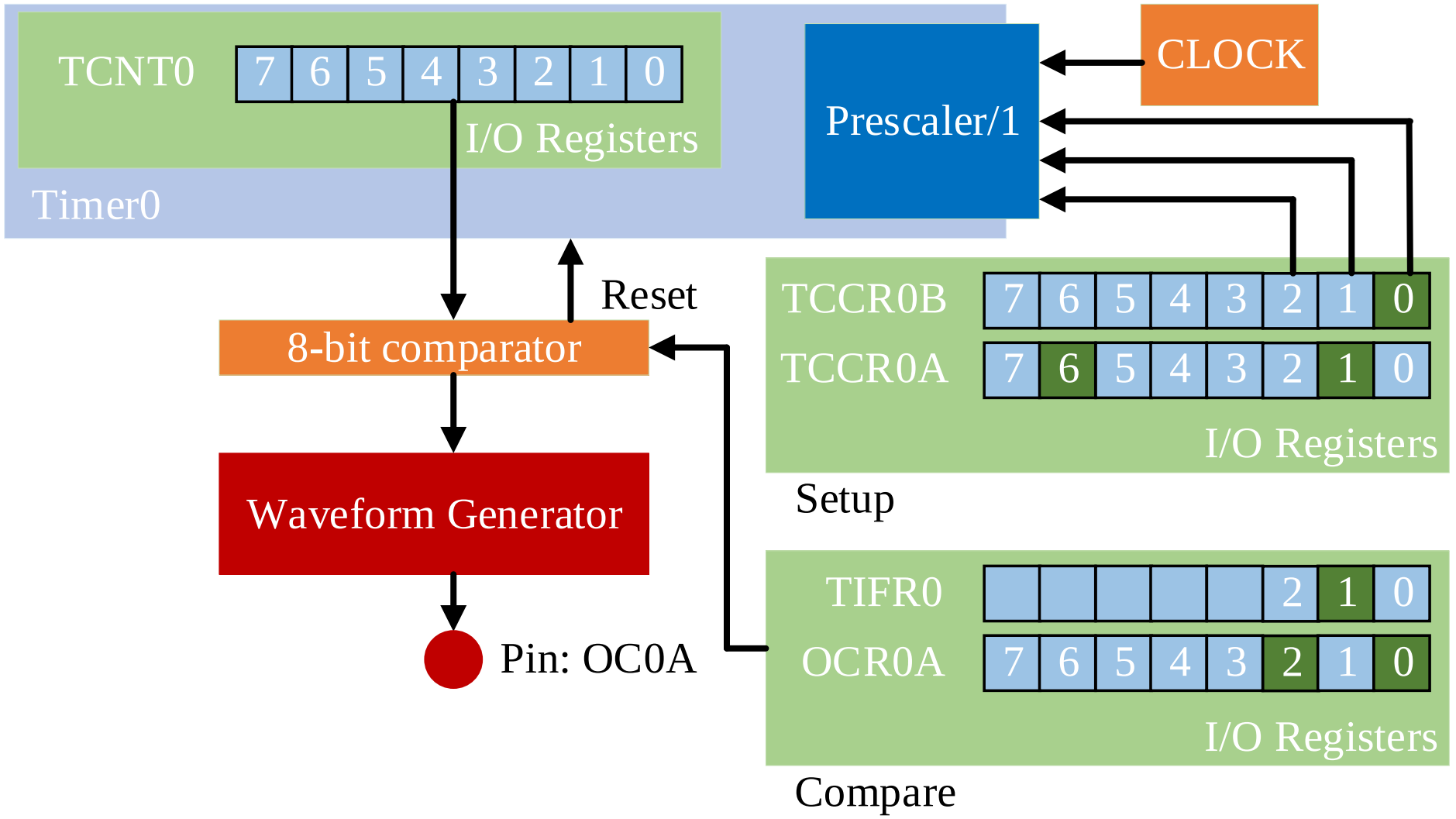
Timing Diagram - II



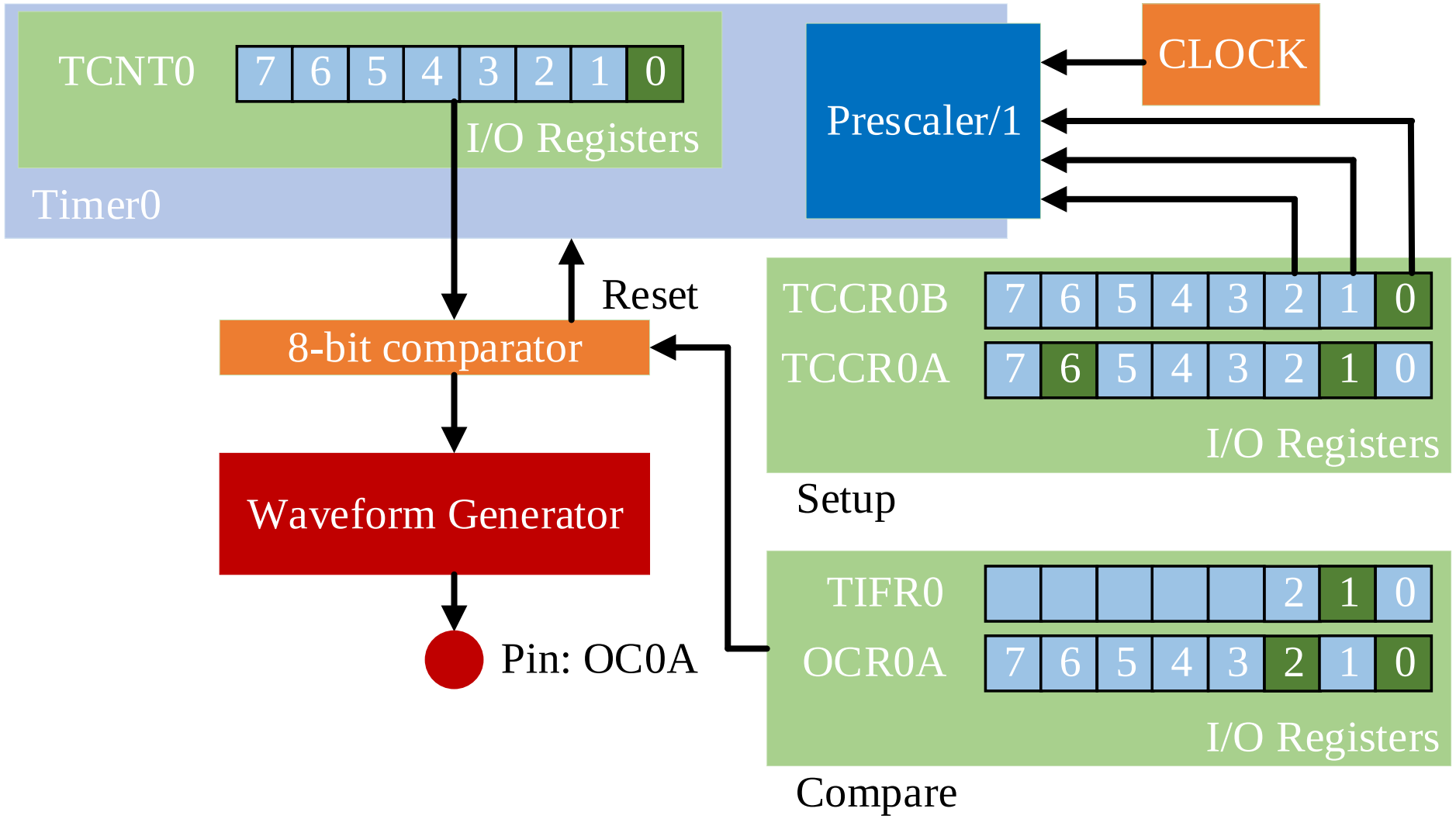
Change OCR0A

- Change the frequency!

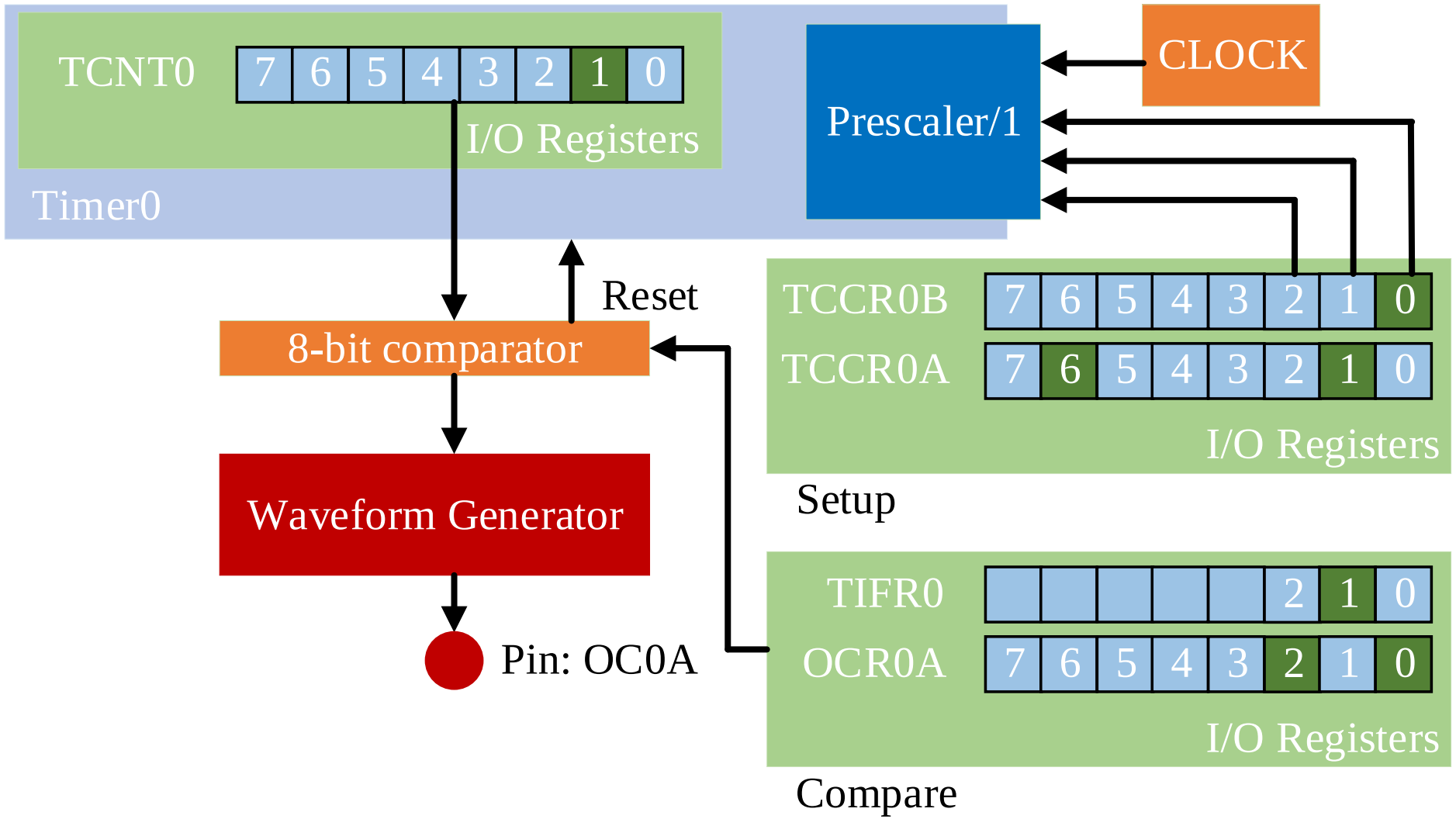
Change OCR0A - I



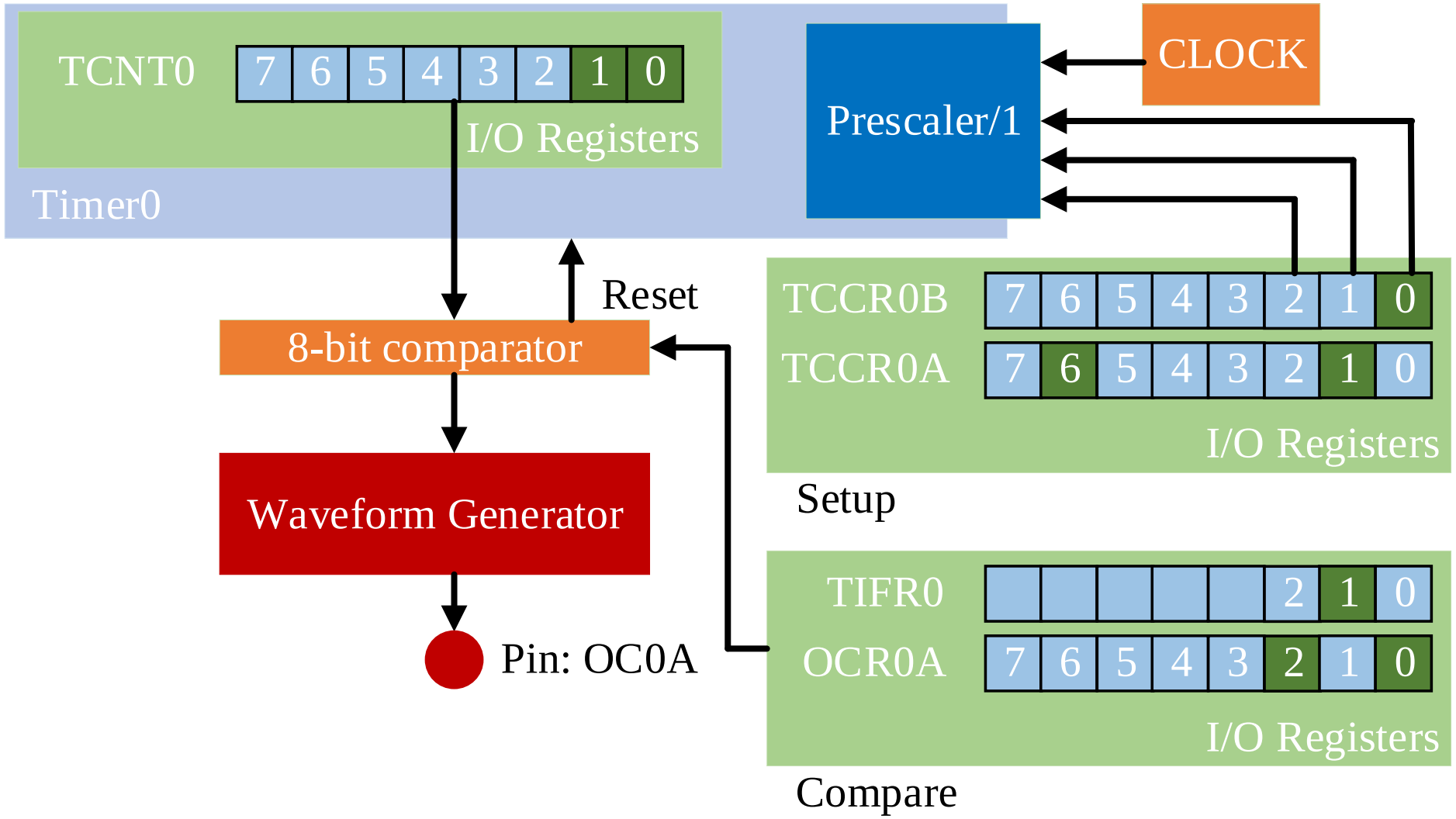
Change OCR0A - II



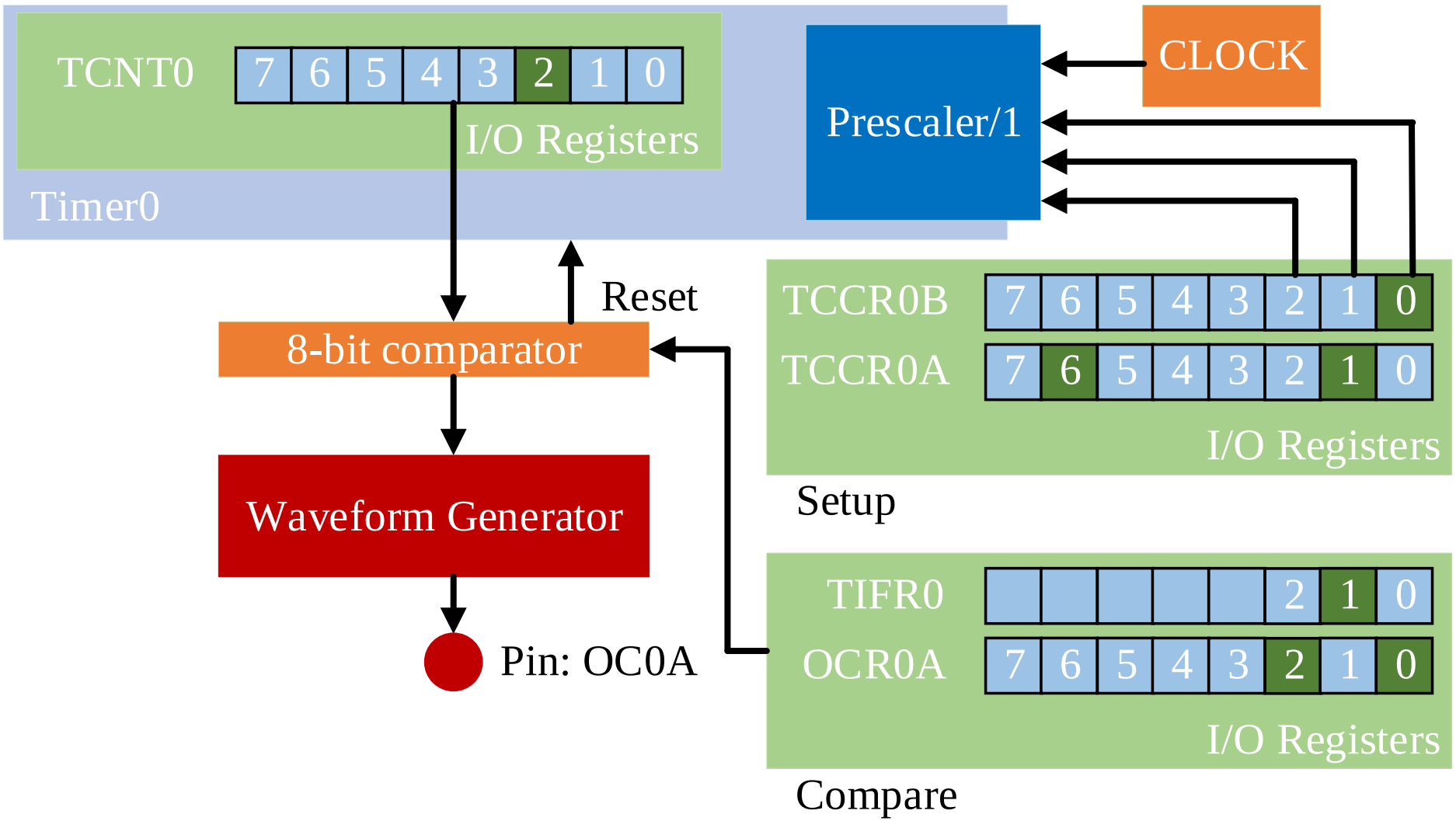
Change OCR0A - III



Change OCR0A - IV



Change OCR0A - V



Change OCR0A - VI

